

# Comprehensive prediction and analysis of human protein essentiality based on a pretrained large language model

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# **Comprehensive prediction and analysis of human protein essentiality based on a pre-trained protein large language model**

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**Supplementary Data 4:** Detailed GO functional enrichment analysis results of differentially essential proteins.

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**Supplementary Data 6:** Detailed statistical analysis results of 10 potential prognostic biomarkers for breast cancer in 8 different cohorts.

**Supplementary Data 7:** Detailed PES scores of all microproteins predicted by PIC models at different levels\*.

**Note:**

\*: This file can be downloaded from Zenodo (<https://zenodo.org/records/13994480>) due to its large size.

**Supplementary Data 8:** Detailed functional enrichment analysis results of top10 microproteins with the highest PES.

## Supplementary Notes

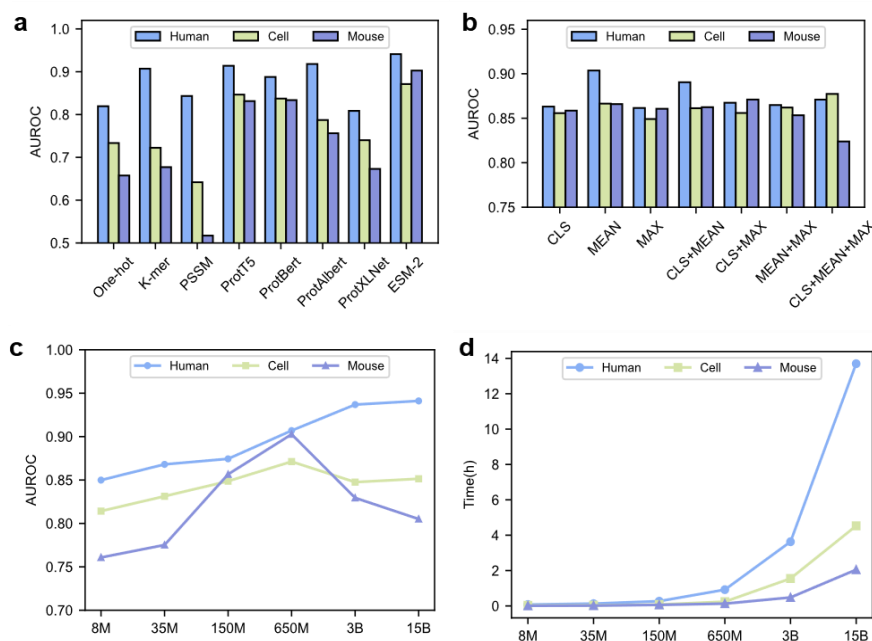
### Supplementary Note 1: Ablation studies of PIC models

The backbone model of PIC is a pre-trained protein language model. Therefore, we first compared the performance of several state-of-the-art PLMs in predicting human protein essentiality, including ESM-2<sup>1</sup>, ProtBert<sup>2</sup>, ProtAlbert<sup>2</sup>, ProtXLNet<sup>2</sup>, and ProtT5<sup>2</sup>. Additionally, we also compared the sequence embeddings generated by PLMs with other types of sequence representations, including One-hot, PSSM, and K-mer. To ensure comprehensive performance comparisons, we evaluated the performance across 325 models at three levels using area under the receiver operating characteristic curve (AUROC) as the evaluation metric. The results indicated that models trained with ESM-2 embeddings achieved the best performance across all three levels compared to other pre-trained PLMs and sequence representation methods (**Fig. 2a and Supplementary Table 10**).

ESM-2 has been pre-trained on over 1.8 billion protein sequences from the UniRef database using self-supervised learning, enabling it to capture intrinsic features within protein sequences<sup>1</sup>. The parameter scale range of the ESM-2 model spans from 8 million to 15 billion, so we compared the performance of ESM-2 models with different parameter sizes, focusing on prediction performance and computer runtime. The results showed that with the increase in model parameters and layers, there was a gradual improvement in model performance. However, the runtime substantially increased as well, especially when the model parameters exceeded 650M. Moreover, an extremely large number of model parameters could lead to overfitting. Specifically, for the mouse-level model with a smaller sample size, the model performance substantially declined when the parameters exceeded 650M (**Fig. 2c, Fig.2d and Supplementary Table 11**). Therefore, to balance model performance and runtime, we finally selected the ESM2\_t33\_650M\_UR50D model with 33 layers and 650M parameters to extract protein sequence embeddings.

The initial embedding generated by ESM-2 is a 1280-dimensional numerical vector for each amino acid in the input protein sequence. Therefore, an optimal pooling method was necessary for feature aggregation to handle protein sequences with varying lengths. We further explored the impact of different feature pooling strategies on model performance. In fact, there are three

mainstream pooling methods for aggregating residue-level features generated by PLMs: CLS (classification token) pooling, Average pooling, and Max pooling. Therefore, we compared the model performance of sequence-level embeddings aggregated by different combinations of these three pooling methods (**Fig. 2b and Supplementary Table 12**). The results showed that embeddings generated using the Average pooling strategy achieved the best performance at both human-level and cell-level, while slightly underperforming the combination of CLS pooling and Max pooling method at the mouse-level. Overall, we chose to use Average pooling method to aggregate features.



**Fig. 2: Ablation studies of PIC models.** (a) The AUROC values for different feature extraction methods are compared across PIC-human, PIC-cell, and PIC-mouse models. Methods include One-hot, K-mer, PSSM, ProtT5, ProtBert, ProtAlbert, ProtXLNet, and ESM-2. (b) The AUROC values for different pooling strategies are compared across PIC-human, PIC-cell, and PIC-mouse models. Strategies include CLS, MEAN, MAX, and their combinations. (c) The AUROC values for PIC-human, PIC-cell, and PIC-mouse models with varying sizes of the model parameters. (d) The training time (in hours) for PIC-human, PIC-cell, and PIC-mouse models with varying sizes of the model parameters. AUROC: Area under the ROC curve. Data for PIC-cell models are presented as mean values in the figure. M: million. B: billion. CLS: using the special [CLS] token's embedding from the output of ESM-2 to represent the entire protein sequence. MEAN: using the average of all token embeddings to represent the entire protein sequence. Max: selecting the maximum value from each dimension of the token embeddings to represent the entire protein sequence.

### Supplementary Note 2: Training details of the PIC models

It is well-known that essential proteins constitute approximately 10% of all human proteins<sup>3</sup>,

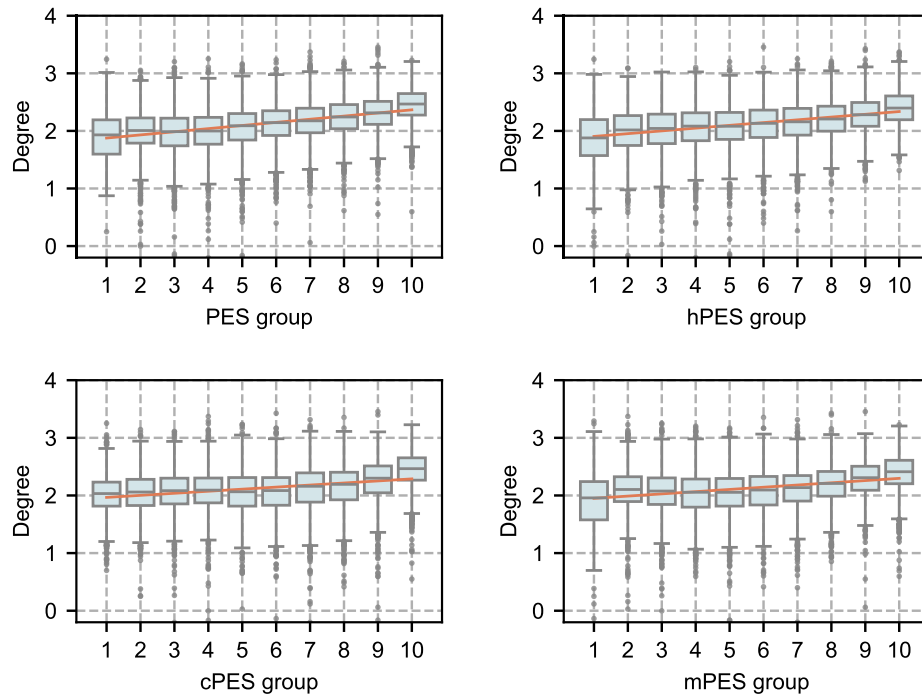
which will lead to highly imbalanced training datasets, which could result in prediction bias during the training process. To address this challenge, we used the `WeightedRandomSampler` function from the Pytorch library to enable the sampling of essential and non-essential proteins with varying weights during the training process. By adjusting these weights, we achieved a relative balanced ratio of positive and negative samples in both the training and the validation sets, thereby alleviating the learning biases during model training. Importantly, we maintained the original proportion of positive and negative samples in the test set to ensure consistency with the real situations. For all PIC models, we partitioned the dataset into 80% for training, 10% for validation, and 10% for testing. To mitigate over-fitting, we incorporated two dropout layers into the architecture: an attention dropout layer and a linear dropout layer. In addition, we implemented a scheduler to dynamically adjust the model's learning rate and employed an early-stopping mechanism to prevent over-fitting. Specifically, When the loss of the validation set is no longer reduced for 2 epochs, the learning rate will be adjusted to 1/10 of the initial learning rate, and when the loss is no longer reduced for 5 epochs, the training will be terminated to prevent over-fitting.

To obtain the optimal hyperparameter combinations, we used the grid search method for each of the 325 PIC models. We first constructed a hyperparameter set, including a total of 256 hyperparameter combinations (**Supplementary Table 13**), and then trained 325 PIC models using different hyperparameter combinations individually. The area under ROC curve (AUROC) was calculated for each of the PIC models on the validation set to evaluate the performance of each hyperparameter combination, allowing us to identify the optimal hyperparameter configuration for each of the 325 PIC models (**Supplementary Table 14 and Supplementary Data 1**).

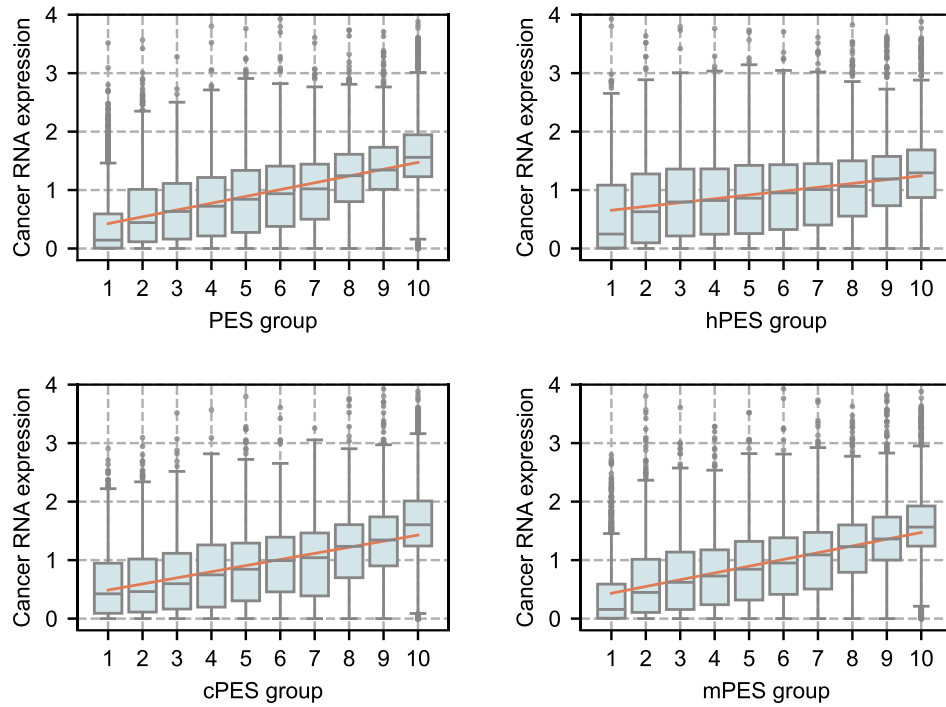
### **Supplementary Note 3: Training devices of the PIC models**

All PIC models in this study were constructed based on Python 3.10.13 and implemented in the Pytorch 1.12.1 library. The training processes were executed on a single Nvidia A100 80GB GPU.

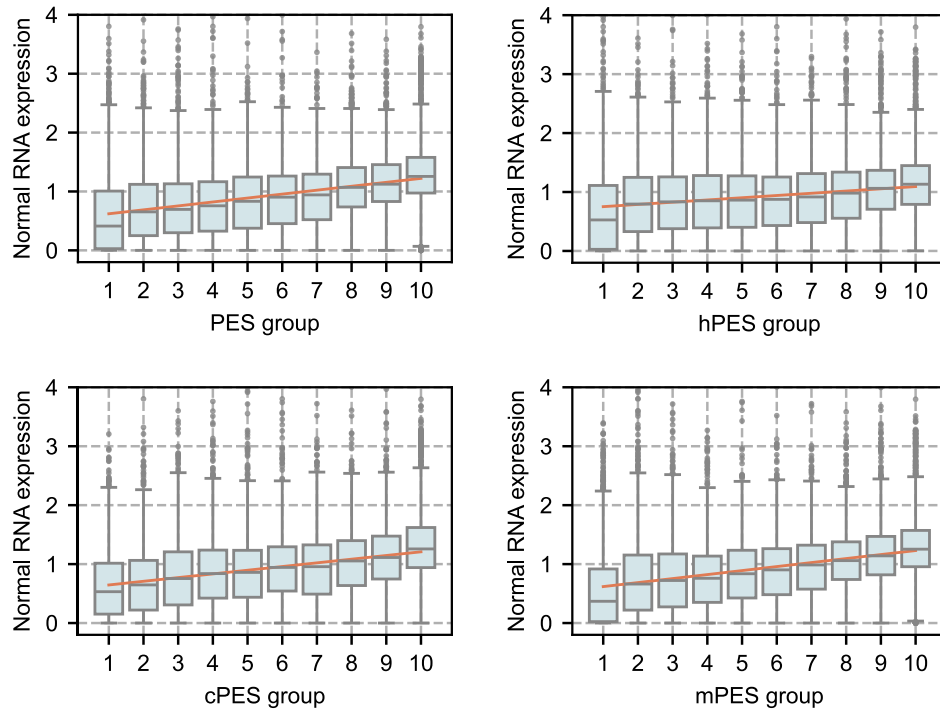
## Supplementary Figures



**Supplementary Figure 1: Correlations between PES and Degree.** Proteins ranked by PES of different levels are divided into ten equal-sized group. Degree: protein degree in protein-protein interaction (PPI) networks. PES: protein essential score. hPES: human-level PES. cPES: cell-level PES. mPES: mouse-level PES. Box plots show the median (center line), the 25th and 75th percentiles (bounds of the box), and the minimum and maximum values (whiskers). The sample size (n) of each subplot is 18726.

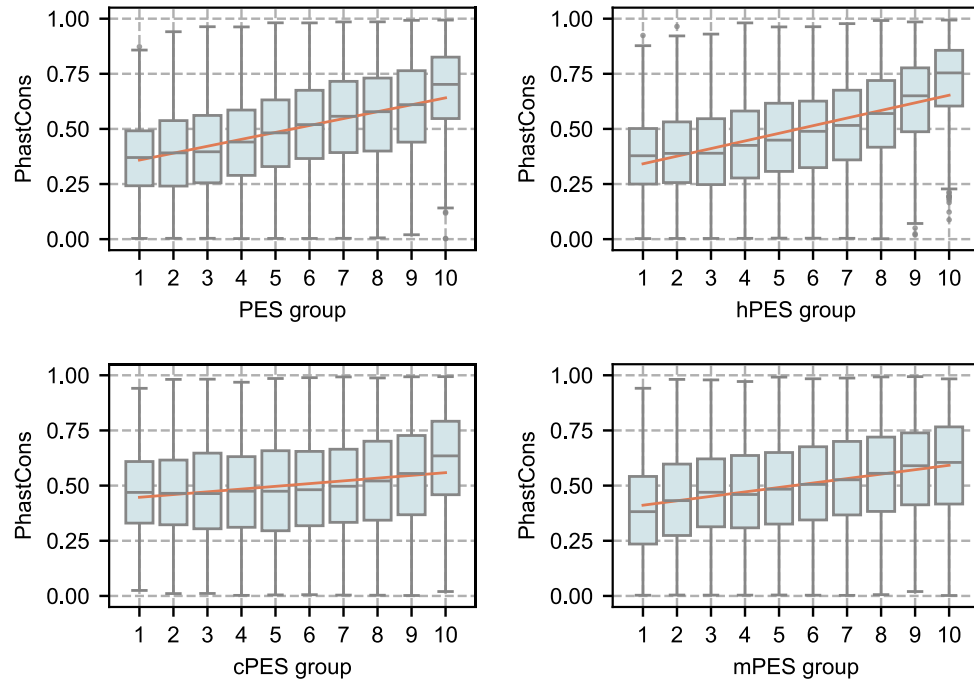


**Supplementary Figure 2: Correlations between PES and Cancer RNA expression.** Proteins ranked by PES of different levels are divided into ten equal-sized group. Cancer RNA expression: Protein expression level in cancer cells. PES: protein essential score. hPES: human-level PES. cPES: cell-level PES. mPES: mouse-level PES. Box plots show the median (center line), the 25th and 75th percentiles (bounds of the box), and the minimum and maximum values (whiskers). The orange line in each box plot represents a linear fit. The sample size (n) of each subplot is 19363.

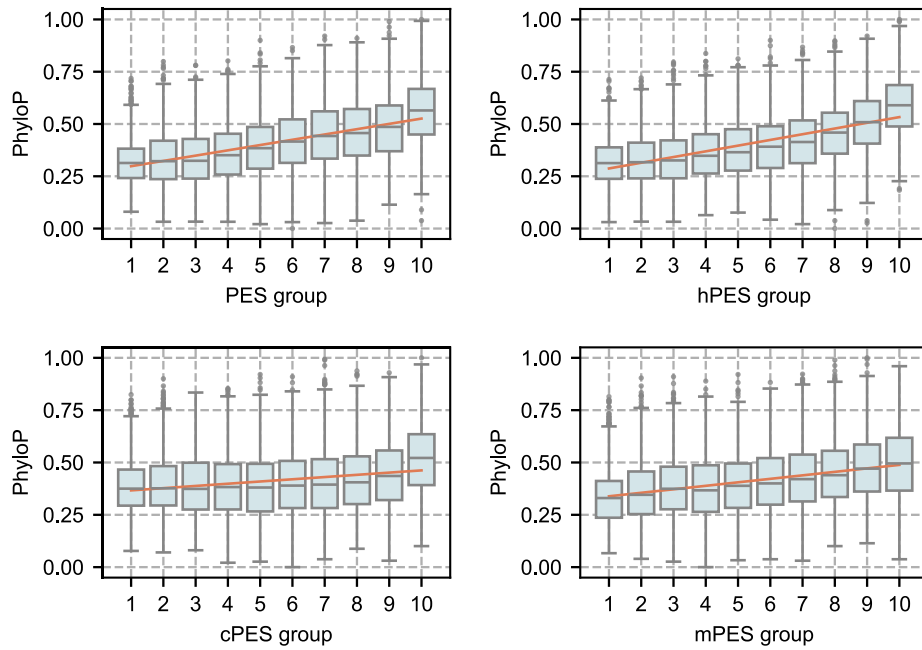


**Supplementary Figure 3: Correlations between PES and Normal RNA expression.**

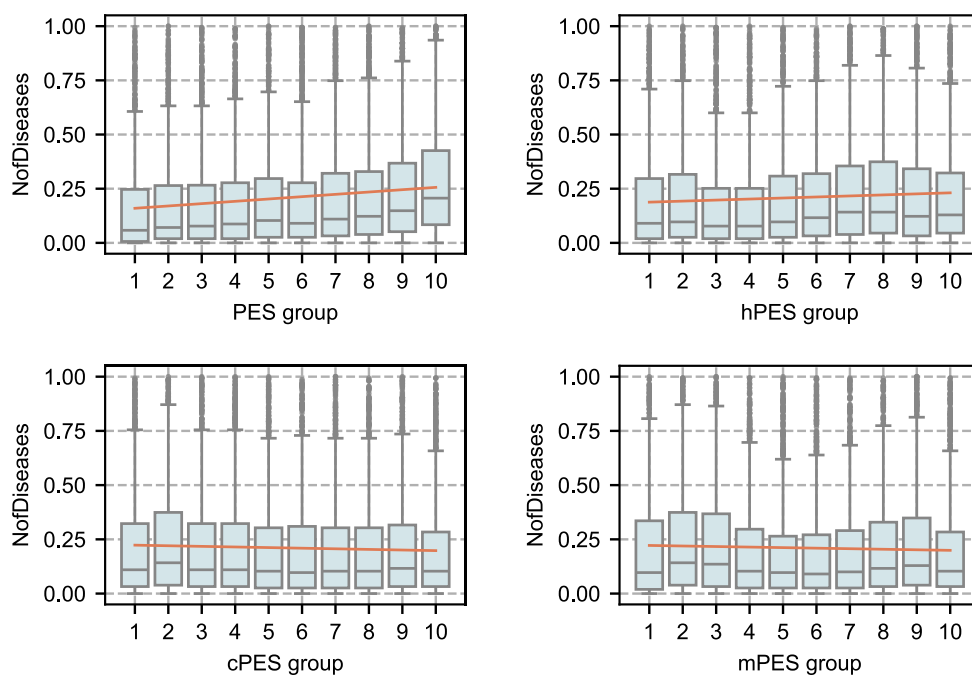
Proteins ranked by PES of different levels are divided into ten equal-sized group. Normal RNA expression: Protein expression level in normal cells. PES: protein essential score. hPES: human-level PES. cPES: cell-level PES. mPES: mouse-level PES. Box plots show the median (center line), the 25th and 75th percentiles (bounds of the box), and the minimum and maximum values (whiskers). The orange line in each box plot represents a linear fit. The sample size (n) of each subplot is 19363.



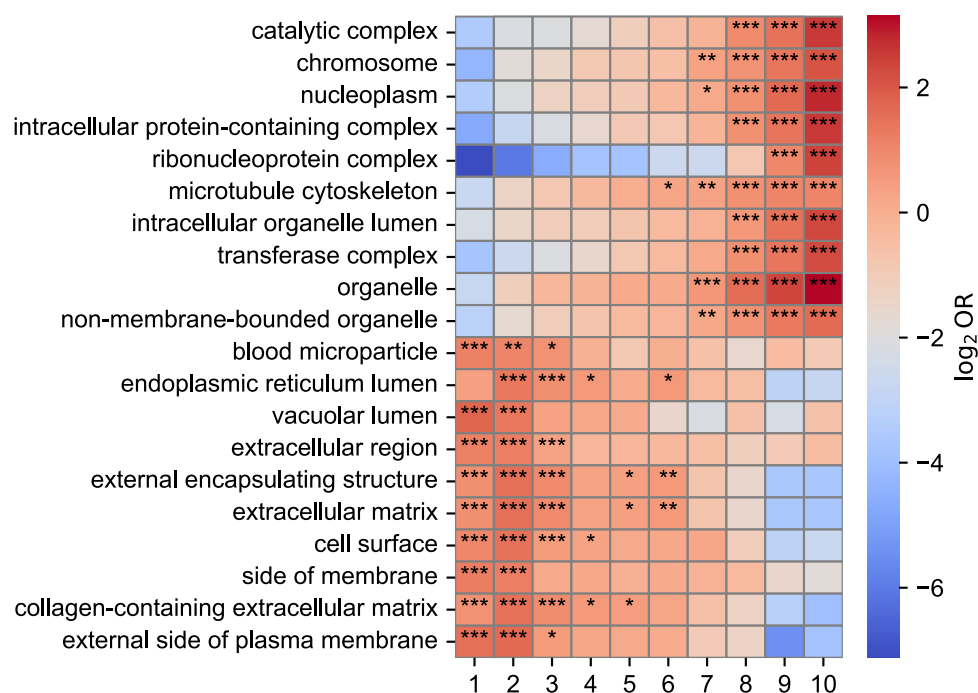
**Supplementary Figure 4: Correlations between PES and PhastCons.** Proteins ranked by PES of different levels are divided into ten equal-sized group. PhastCons: Conservation score across species. PES: protein essential score. hPES: human-level PES. cPES: cell-level PES. mPES: mouse-level PES. Box plots show the median (center line), the 25th and 75th percentiles (bounds of the box), and the minimum and maximum values (whiskers). The orange line in each box plot represents a linear fit. The sample size (n) of each subplot is 16437.



**Supplementary Figure 5: Correlations between PES and PhyloP.** Proteins ranked by PES of different levels are divided into ten equal-sized group. PhyloP: Phylogenetic p-value indicating evolutionary conservation. PES: protein essential score. hPES: human-level PES. cPES: cell-level PES. mPES: mouse-level PES. Box plots show the median (center line), the 25th and 75th percentiles (bounds of the box), and the minimum and maximum values (whiskers). The orange line in each box plot represents a linear fit. The sample size (n) of each subplot is 16437.



**Supplementary Figure 6: Correlations between PES and NofDiseases.** Proteins ranked by PES of different levels are divided into ten equal-sized group. NofDiseases: Number of diseases related to a certain protein. PES: protein essential score. hPES: human-level PES. cPES: cell-level PES. mPES: mouse-level PES. Box plots show the median (center line), the 25th and 75th percentiles (bounds of the box), and the minimum and maximum values (whiskers). The orange line in each box plot represents a linear fit. The sample size (n) of each subplot is 15190.



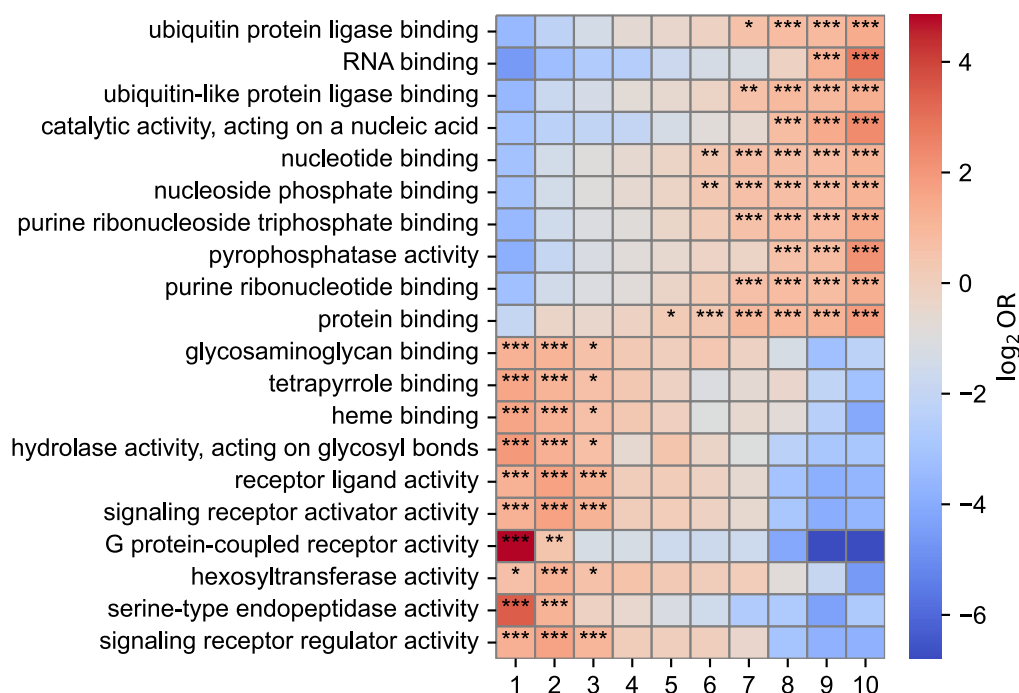
**Supplementary Figure 7: CC enrichment analysis results of proteins with different PES.**

**Proteins ranked by PES were divided into ten equal-sized group. CC: Cellular Component.**

Log<sub>2</sub>OR: Logarithm of the odds ratio value of each term. \* refers to p<0.05; \*\* refers to p<0.01;

\*\*\* refers to p<0.001. P-value was calculated using Fisher's exact test (one-sided, n=18870,

where n represents the total number of genes).



**Supplementary Figure 8: MF enrichment analysis results of proteins with different PES.**

**Proteins ranked by PES were divided into ten equal-sized group.** MF: Molecular Function.

Log<sub>2</sub>OR: Logarithm of the odds ratio value of each term. \* refers to p<0.05; \*\* refers to p<0.01;

\*\*\* refers to p<0.001. P-value was calculated using Fisher's exact test (one-sided, n=18870,

where n represents the total number of genes).

## Supplementary Tables

**Supplementary Table 1. Detailed performance results of 28 disease-level PIC models.**

| Disease                            | Num. of cell line | AUROC range   |
|------------------------------------|-------------------|---------------|
| Acute Myeloid Leukemia             | 2                 | 0.8798-0.9029 |
| B-Cell Non-Hodgkin's Lymphoma      | 5                 | 0.8248-0.8770 |
| Biliary Tract Carcinoma            | 1                 | 0.8276-0.8276 |
| Breast Carcinoma                   | 25                | 0.7783-0.8983 |
| Colorectal Carcinoma               | 30                | 0.7949-0.8894 |
| Endometrial Carcinoma              | 7                 | 0.8424-0.8883 |
| Esophageal Carcinoma               | 7                 | 0.8438-0.8773 |
| Esophageal Squamous Cell Carcinoma | 19                | 0.8449-0.8860 |
| Ewing's Sarcoma                    | 10                | 0.8352-0.8824 |
| Gastric Carcinoma                  | 14                | 0.7663-0.8870 |
| Glioblastoma                       | 24                | 0.8216-0.8951 |
| Glioma                             | 9                 | 0.8279-0.8874 |
| Head and Neck Carcinoma            | 16                | 0.8355-0.8965 |
| Kidney Carcinoma                   | 3                 | 0.8465-0.8507 |
| Melanoma                           | 4                 | 0.8271-0.8720 |
| Mesothelioma                       | 1                 | 0.8656-0.8656 |
| Neuroblastoma                      | 17                | 0.7543-0.8838 |
| Non-Cancerous                      | 3                 | 0.8394-0.8548 |
| Non-Small Cell Lung Carcinoma      | 28                | 0.7938-0.8915 |
| Oral Cavity Carcinoma              | 18                | 0.8436-0.8958 |
| Osteosarcoma                       | 2                 | 0.8333-0.8379 |
| Other Sarcomas                     | 2                 | 0.8346-0.8672 |
| Other Solid Cancers                | 2                 | 0.8614-0.8763 |
| Ovarian Carcinoma                  | 31                | 0.7564-0.8989 |
| Pancreatic Carcinoma               | 23                | 0.8061-0.8843 |
| Plasma Cell Myeloma                | 5                 | 0.8284-0.8879 |
| Prostate Carcinoma                 | 3                 | 0.8275-0.8595 |
| Squamous Cell Lung Carcinoma       | 12                | 0.8347-0.8799 |

**Supplementary Table 2. Detailed performance results of 19 tissue-level PIC models.**

| <b>Tissue</b>               | <b>Num. of cell line</b> | <b>AUROC Range</b> |
|-----------------------------|--------------------------|--------------------|
| Biliary Tract               | 1                        | 0.8276-0.8276      |
| Bone                        | 12                       | 0.8333-0.8824      |
| Breast                      | 25                       | 0.7783-0.8983      |
| Central Nervous System      | 34                       | 0.8216-0.8951      |
| Endometrium                 | 8                        | 0.8424-0.8883      |
| Esophagus                   | 26                       | 0.8438-0.8860      |
| Eye                         | 1                        | 0.8477-0.8477      |
| Haematopoietic and Lymphoid | 14                       | 0.8248-0.9029      |
| Head and Neck               | 34                       | 0.8355-0.8965      |
| Kidney                      | 3                        | 0.8465-0.8507      |
| Large Intestine             | 30                       | 0.7949-0.8894      |
| Lung                        | 42                       | 0.7938-0.8915      |
| Ovary                       | 31                       | 0.7564-0.8989      |
| Pancreas                    | 23                       | 0.8061-0.8843      |
| Peripheral Nervous System   | 17                       | 0.7543-0.8838      |
| Prostate                    | 3                        | 0.8275-0.8595      |
| Skin                        | 4                        | 0.8271-0.8720      |
| Soft Tissue                 | 1                        | 0.8346-0.8346      |
| Stomach                     | 14                       | 0.7663-0.8870      |

**Supplementary Table 3. Overall performance of PIC models at three levels on the independent test sets.** The independent test sets were extreme imbalanced. The performance of the PIC-cell model was the median value of 323 cell-level PIC models.

| Model     | Accuracy | Recall | Precision | F1     | AUROC  | AUPRC  |
|-----------|----------|--------|-----------|--------|--------|--------|
| PIC-human | 0.8390   | 0.8324 | 0.5924    | 0.6921 | 0.9132 | 0.7864 |
| PIC-cell  | 0.8231   | 0.7135 | 0.2674    | 0.3867 | 0.8572 | 0.4277 |
| PIC-mouse | 0.8437   | 0.7045 | 0.2672    | 0.3875 | 0.8736 | 0.3064 |

**Supplementary Table 4. Detailed performance of the 323 cell-line level PIC models.**

| Cell line   | Accuracy | AUROC  | Recall | AUPRC  | F1     | Loss   |
|-------------|----------|--------|--------|--------|--------|--------|
| RERF-GC-1B  | 0.8323   | 0.8286 | 0.6667 | 0.3739 | 0.3514 | 0.3844 |
| A172        | 0.8544   | 0.8701 | 0.7542 | 0.4621 | 0.4354 | 0.3584 |
| KP-1N       | 0.82     | 0.8315 | 0.6829 | 0.4632 | 0.4203 | 0.3988 |
| D-423MG     | 0.8084   | 0.8384 | 0.6989 | 0.4642 | 0.4414 | 0.4156 |
| HCE-4       | 0.8149   | 0.8489 | 0.6724 | 0.3598 | 0.3291 | 0.3776 |
| MOG-G-UVW   | 0.8177   | 0.8657 | 0.7208 | 0.4569 | 0.415  | 0.4015 |
| IGROV-1     | 0.8161   | 0.8161 | 0.6452 | 0.3946 | 0.3361 | 0.3964 |
| SCC-4       | 0.8166   | 0.8581 | 0.7902 | 0.4225 | 0.4177 | 0.4024 |
| HCC-15      | 0.9284   | 0.8487 | 0.4571 | 0.2557 | 0.2065 | 0.1822 |
| LB771-HNC   | 0.8388   | 0.8542 | 0.7179 | 0.4818 | 0.4471 | 0.3578 |
| Caov-4      | 0.826    | 0.854  | 0.6752 | 0.3726 | 0.3457 | 0.3665 |
| SF126       | 0.8388   | 0.8658 | 0.6687 | 0.4876 | 0.4404 | 0.3573 |
| PCI-30      | 0.8376   | 0.8355 | 0.6566 | 0.4637 | 0.4386 | 0.3778 |
| NCI-H226    | 0.8347   | 0.8656 | 0.7222 | 0.339  | 0.314  | 0.3602 |
| 22RV1       | 0.8489   | 0.8275 | 0.5579 | 0.2753 | 0.2723 | 0.3291 |
| PA-TU-8988T | 0.8265   | 0.8843 | 0.7848 | 0.4512 | 0.4542 | 0.3756 |
| HSC-3       | 0.8132   | 0.8552 | 0.7372 | 0.4511 | 0.4174 | 0.3829 |
| SK-N-SH     | 0.8415   | 0.8838 | 0.7672 | 0.4029 | 0.3428 | 0.3487 |
| KYSE-450    | 0.8417   | 0.8708 | 0.7324 | 0.4411 | 0.4333 | 0.3606 |
| SW1573      | 0.8157   | 0.8721 | 0.7597 | 0.5078 | 0.3977 | 0.3749 |
| BB30-HNC    | 0.8195   | 0.8696 | 0.7389 | 0.4857 | 0.4049 | 0.3871 |
| 23132-87    | 0.8277   | 0.8608 | 0.6796 | 0.4006 | 0.4001 | 0.3897 |
| OE33        | 0.8241   | 0.8773 | 0.7705 | 0.4834 | 0.3837 | 0.3751 |
| MKN28       | 0.8126   | 0.8769 | 0.7866 | 0.4767 | 0.4448 | 0.4043 |
| UWB1.289    | 0.7951   | 0.8772 | 0.7473 | 0.4999 | 0.3549 | 0.4206 |
| KP-N-YN     | 0.8027   | 0.8512 | 0.7143 | 0.4295 | 0.3825 | 0.4104 |
| EVSA-T      | 0.8236   | 0.8666 | 0.7075 | 0.4608 | 0.407  | 0.3787 |
| SNU-16      | 0.8028   | 0.8126 | 0.6695 | 0.3787 | 0.3641 | 0.4252 |
| SK-N-AS     | 0.8008   | 0.8741 | 0.7912 | 0.5409 | 0.4571 | 0.4231 |

|            |        |        |        |        |        |        |
|------------|--------|--------|--------|--------|--------|--------|
| Becker     | 0.8101 | 0.8717 | 0.7521 | 0.4944 | 0.4169 | 0.3918 |
| EC-GI-10   | 0.8329 | 0.8715 | 0.7078 | 0.4602 | 0.4317 | 0.3791 |
| SNU-81     | 0.8243 | 0.8368 | 0.6367 | 0.4533 | 0.3683 | 0.3741 |
| OE21       | 0.819  | 0.8562 | 0.7207 | 0.5253 | 0.4534 | 0.3858 |
| GAMG       | 0.8073 | 0.854  | 0.75   | 0.4909 | 0.4547 | 0.4235 |
| EFO-27     | 0.8434 | 0.8436 | 0.6159 | 0.4446 | 0.3872 | 0.3558 |
| RKN        | 0.8468 | 0.8346 | 0.625  | 0.4226 | 0.3926 | 0.3482 |
| CAL-27     | 0.7905 | 0.8675 | 0.8278 | 0.4965 | 0.4529 | 0.4184 |
| KYSE-520   | 0.8225 | 0.8493 | 0.7109 | 0.4045 | 0.3737 | 0.3757 |
| HCC1937    | 0.8229 | 0.8687 | 0.7368 | 0.4272 | 0.3559 | 0.3761 |
| COLO-684   | 0.8101 | 0.8656 | 0.7386 | 0.4759 | 0.4094 | 0.415  |
| NB10       | 0.8462 | 0.8751 | 0.6988 | 0.3373 | 0.3053 | 0.3397 |
| A2780cis   | 0.825  | 0.8749 | 0.6736 | 0.433  | 0.3703 | 0.3627 |
| NCI-H1355  | 0.812  | 0.8832 | 0.7813 | 0.4917 | 0.3824 | 0.396  |
| NCI-H322M  | 0.8323 | 0.7994 | 0.5773 | 0.297  | 0.28   | 0.3828 |
| MCAS       | 0.8649 | 0.8547 | 0.549  | 0.2515 | 0.1944 | 0.3024 |
| HO-1-u-1   | 0.8341 | 0.8855 | 0.7829 | 0.4531 | 0.4551 | 0.3715 |
| NCI-H3122  | 0.8171 | 0.7938 | 0.5667 | 0.3183 | 0.3022 | 0.3942 |
| SW620      | 0.8193 | 0.8742 | 0.7822 | 0.4631 | 0.3846 | 0.3913 |
| NCI-H2170  | 0.7903 | 0.8347 | 0.7237 | 0.3656 | 0.3793 | 0.4378 |
| 8-MG-BA    | 0.792  | 0.8481 | 0.7245 | 0.4619 | 0.4339 | 0.4405 |
| SK-MG-1    | 0.8178 | 0.8874 | 0.7866 | 0.4672 | 0.4518 | 0.3852 |
| DOV13      | 0.8062 | 0.8559 | 0.764  | 0.4119 | 0.4249 | 0.4221 |
| SKN-3      | 0.8272 | 0.8699 | 0.7103 | 0.3358 | 0.3049 | 0.3614 |
| U-87-MG    | 0.871  | 0.8698 | 0.631  | 0.469  | 0.4115 | 0.3031 |
| SAS        | 0.8073 | 0.8595 | 0.6887 | 0.4431 | 0.3859 | 0.3964 |
| DOK        | 0.8213 | 0.8882 | 0.7845 | 0.4755 | 0.3722 | 0.3649 |
| NB6        | 0.8027 | 0.8607 | 0.7432 | 0.3983 | 0.3936 | 0.4237 |
| IST-MEL1   | 0.8289 | 0.8524 | 0.7    | 0.4267 | 0.3636 | 0.3862 |
| T47D       | 0.8368 | 0.8864 | 0.7343 | 0.4618 | 0.3714 | 0.337  |
| PCI-6A     | 0.8184 | 0.8717 | 0.7456 | 0.3599 | 0.3527 | 0.3751 |
| LS-123     | 0.8475 | 0.8504 | 0.587  | 0.231  | 0.1709 | 0.3392 |
| GI-1       | 0.8002 | 0.8476 | 0.7568 | 0.4628 | 0.4494 | 0.4435 |
| NCI-H2023  | 0.8254 | 0.8799 | 0.7325 | 0.4594 | 0.434  | 0.3664 |
| SW837      | 0.8242 | 0.8696 | 0.7512 | 0.4875 | 0.3859 | 0.374  |
| OCUB-M     | 0.8283 | 0.8876 | 0.8065 | 0.4069 | 0.404  | 0.3804 |
| MC-IXC     | 0.7816 | 0.8352 | 0.7688 | 0.4647 | 0.4493 | 0.4648 |
| CAL-72     | 0.8393 | 0.8333 | 0.5833 | 0.2631 | 0.262  | 0.344  |
| BE2-M17    | 0.8606 | 0.8742 | 0.7072 | 0.5088 | 0.4709 | 0.3318 |
| PANC-08-13 | 0.8196 | 0.8414 | 0.687  | 0.3256 | 0.3376 | 0.377  |
| SW1990     | 0.8228 | 0.8498 | 0.6866 | 0.4627 | 0.399  | 0.3845 |
| TGW        | 0.8064 | 0.8566 | 0.7579 | 0.4745 | 0.416  | 0.4122 |
| PEO4       | 0.8271 | 0.8657 | 0.7094 | 0.4279 | 0.3585 | 0.3886 |

|             |        |        |        |        |        |        |
|-------------|--------|--------|--------|--------|--------|--------|
| HPAF-II     | 0.8328 | 0.8593 | 0.656  | 0.4033 | 0.3636 | 0.3593 |
| KM12        | 0.8369 | 0.8561 | 0.7288 | 0.4347 | 0.3805 | 0.3517 |
| HCC-78      | 0.7967 | 0.8575 | 0.74   | 0.38   | 0.2978 | 0.4137 |
| FLO-1       | 0.812  | 0.8466 | 0.74   | 0.3962 | 0.4073 | 0.4088 |
| BICR78      | 0.8267 | 0.8436 | 0.69   | 0.3163 | 0.3024 | 0.3834 |
| SJSA-1      | 0.8353 | 0.8379 | 0.6865 | 0.4769 | 0.473  | 0.3798 |
| VMRC-LCD    | 0.8108 | 0.8805 | 0.7763 | 0.5241 | 0.4257 | 0.397  |
| SK-GT-4     | 0.8352 | 0.8671 | 0.7305 | 0.4338 | 0.4213 | 0.3624 |
| KP-3        | 0.8399 | 0.8427 | 0.688  | 0.3849 | 0.3848 | 0.3726 |
| SU-DHL-10   | 0.8213 | 0.8654 | 0.7464 | 0.4299 | 0.3989 | 0.393  |
| SU-DHL-8    | 0.8015 | 0.877  | 0.7515 | 0.4579 | 0.4232 | 0.3979 |
| OPM-2       | 0.837  | 0.8284 | 0.6111 | 0.3873 | 0.386  | 0.3634 |
| BICR22      | 0.83   | 0.8952 | 0.7512 | 0.4593 | 0.3745 | 0.3525 |
| NB69        | 0.8218 | 0.8539 | 0.7417 | 0.4519 | 0.4226 | 0.3828 |
| JHU-022     | 0.8161 | 0.8737 | 0.7424 | 0.4537 | 0.3828 | 0.3738 |
| SCH         | 0.8306 | 0.7663 | 0.5658 | 0.2566 | 0.2281 | 0.3729 |
| hTERT-RPE-1 | 0.7986 | 0.8477 | 0.7273 | 0.3995 | 0.3372 | 0.4145 |
| SUIT-2      | 0.8273 | 0.8546 | 0.7222 | 0.5066 | 0.418  | 0.3778 |
| JIMT-1      | 0.8299 | 0.846  | 0.7391 | 0.3431 | 0.368  | 0.3847 |
| NCI-H1993   | 0.7829 | 0.8221 | 0.7183 | 0.4025 | 0.3536 | 0.4615 |
| PANC-04-03  | 0.8271 | 0.8115 | 0.5741 | 0.3981 | 0.3851 | 0.3873 |
| LOU-NH91    | 0.8148 | 0.8403 | 0.6857 | 0.4465 | 0.3765 | 0.3932 |
| OCI-AML3    | 0.8504 | 0.9029 | 0.7922 | 0.5652 | 0.487  | 0.3358 |
| TE-5        | 0.8244 | 0.8649 | 0.7659 | 0.4462 | 0.3978 | 0.3925 |
| COR-L23     | 0.8165 | 0.8654 | 0.7397 | 0.4681 | 0.4068 | 0.3765 |
| NB13        | 0.7769 | 0.7543 | 0.5894 | 0.3276 | 0.3173 | 0.4616 |
| GP5d        | 0.8405 | 0.8701 | 0.7143 | 0.4987 | 0.3687 | 0.338  |
| HEC-1       | 0.7734 | 0.8424 | 0.7014 | 0.4541 | 0.3418 | 0.4476 |
| COLO-678    | 0.9069 | 0.7949 | 0.4359 | 0.1451 | 0.1753 | 0.2418 |
| no-11       | 0.8184 | 0.8216 | 0.712  | 0.4095 | 0.3633 | 0.4149 |
| HCC1954     | 0.8108 | 0.8439 | 0.7019 | 0.3423 | 0.31   | 0.3899 |
| 42-MG-BA    | 0.8106 | 0.8537 | 0.6907 | 0.366  | 0.3183 | 0.3981 |
| GB-1        | 0.8206 | 0.8624 | 0.7385 | 0.4544 | 0.384  | 0.3871 |
| KURAMOCHI   | 0.8218 | 0.8734 | 0.7681 | 0.4266 | 0.4093 | 0.3906 |
| RCM-1       | 0.7863 | 0.8331 | 0.7131 | 0.3524 | 0.3216 | 0.4412 |
| EW-1        | 0.8224 | 0.8824 | 0.7625 | 0.464  | 0.4444 | 0.3701 |
| BHY         | 0.8071 | 0.8623 | 0.7499 | 0.4537 | 0.3901 | 0.4134 |
| ESO51       | 0.8155 | 0.8438 | 0.6885 | 0.3511 | 0.3464 | 0.4122 |
| L-363       | 0.837  | 0.8879 | 0.7481 | 0.4799 | 0.4118 | 0.3427 |
| NCI-H1915   | 0.8364 | 0.817  | 0.6404 | 0.3425 | 0.3419 | 0.3697 |
| TE-8        | 0.8176 | 0.8576 | 0.7327 | 0.5553 | 0.4532 | 0.3974 |
| AU565       | 0.8141 | 0.8767 | 0.7548 | 0.4392 | 0.3462 | 0.3854 |
| KNS-42      | 0.8317 | 0.8611 | 0.75   | 0.4629 | 0.4138 | 0.3869 |

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|------------|--------|--------|--------|--------|--------|--------|
| MDA-MB-436 | 0.816  | 0.8385 | 0.6985 | 0.386  | 0.3755 | 0.4144 |
| LP-1       | 0.8079 | 0.8702 | 0.75   | 0.4492 | 0.3889 | 0.3975 |
| NCI-H358   | 0.8614 | 0.8496 | 0.6744 | 0.3268 | 0.3277 | 0.318  |
| PCI-4B     | 0.8277 | 0.88   | 0.7639 | 0.4906 | 0.4264 | 0.3652 |
| OVCA420    | 0.8259 | 0.8629 | 0.7209 | 0.3835 | 0.3835 | 0.3622 |
| AGS        | 0.8204 | 0.8694 | 0.7176 | 0.4878 | 0.4422 | 0.3829 |
| KYSE-140   | 0.8393 | 0.871  | 0.7055 | 0.4842 | 0.4274 | 0.3534 |
| NUGC-3     | 0.8021 | 0.8487 | 0.747  | 0.4585 | 0.4218 | 0.412  |
| JHOS-2     | 0.8318 | 0.8886 | 0.7881 | 0.5208 | 0.4516 | 0.3506 |
| NB5        | 0.8463 | 0.86   | 0.6776 | 0.431  | 0.4383 | 0.345  |
| HOP-62     | 0.8236 | 0.8579 | 0.6667 | 0.3012 | 0.2904 | 0.3703 |
| NMC-G1     | 0.8737 | 0.8584 | 0.6765 | 0.3574 | 0.2977 | 0.2887 |
| ES4        | 0.8609 | 0.861  | 0.766  | 0.3743 | 0.376  | 0.3181 |
| SNU-C1     | 0.8316 | 0.8468 | 0.6    | 0.3697 | 0.3193 | 0.3704 |
| NCI-H23    | 0.8102 | 0.8598 | 0.7371 | 0.4954 | 0.4418 | 0.4061 |
| U251       | 0.8199 | 0.8733 | 0.7536 | 0.4687 | 0.4088 | 0.3782 |
| SW48       | 0.8114 | 0.8632 | 0.726  | 0.4947 | 0.3955 | 0.3811 |
| MIA-PaCa-2 | 0.8382 | 0.8585 | 0.7059 | 0.445  | 0.4372 | 0.3653 |
| CAS-1      | 0.8468 | 0.8807 | 0.7103 | 0.3557 | 0.3663 | 0.3423 |
| NCI-H2087  | 0.8259 | 0.8559 | 0.6928 | 0.4247 | 0.4149 | 0.3882 |
| PEO1       | 0.812  | 0.8789 | 0.7939 | 0.552  | 0.4479 | 0.3995 |
| Hs-578-T   | 0.8439 | 0.8539 | 0.7064 | 0.3507 | 0.3649 | 0.3504 |
| SK-N-DZ    | 0.8235 | 0.7966 | 0.5094 | 0.2436 | 0.2628 | 0.3832 |
| ES5        | 0.8101 | 0.8669 | 0.7546 | 0.5145 | 0.4301 | 0.3921 |
| TE-10      | 0.8101 | 0.8529 | 0.6937 | 0.4614 | 0.3281 | 0.3967 |
| AsPC-1     | 0.842  | 0.8734 | 0.7586 | 0.4506 | 0.3986 | 0.3723 |
| PANC-10-05 | 0.8416 | 0.8433 | 0.6636 | 0.3262 | 0.343  | 0.3491 |
| SNG-M      | 0.8053 | 0.8602 | 0.7272 | 0.4399 | 0.4117 | 0.4067 |
| JHU-029    | 0.8323 | 0.8465 | 0.6608 | 0.4366 | 0.4397 | 0.3757 |
| LC-1-sq    | 0.8242 | 0.8677 | 0.7807 | 0.4545 | 0.3708 | 0.3772 |
| HCC70      | 0.8475 | 0.8983 | 0.7727 | 0.3776 | 0.3417 | 0.3186 |
| MFE-280    | 0.8328 | 0.8452 | 0.6437 | 0.3397 | 0.2807 | 0.3631 |
| DU-145     | 0.8201 | 0.8575 | 0.6879 | 0.4462 | 0.4114 | 0.3857 |
| TC-71      | 0.826  | 0.8611 | 0.709  | 0.5027 | 0.4692 | 0.3856 |
| LN-229     | 0.83   | 0.8564 | 0.7386 | 0.47   | 0.4363 | 0.3835 |
| A253       | 0.7761 | 0.8579 | 0.7957 | 0.3999 | 0.326  | 0.4418 |
| KYSE-510   | 0.8231 | 0.8706 | 0.7572 | 0.4904 | 0.4629 | 0.3975 |
| ES8        | 0.8516 | 0.8485 | 0.6508 | 0.3734 | 0.3914 | 0.3415 |
| SK-MES-1   | 0.8229 | 0.8655 | 0.7622 | 0.4724 | 0.4176 | 0.3926 |
| KON        | 0.8143 | 0.8545 | 0.7235 | 0.4588 | 0.4354 | 0.3959 |
| HCC1187    | 0.813  | 0.8216 | 0.656  | 0.397  | 0.3381 | 0.4001 |
| ARH-77     | 0.8214 | 0.8394 | 0.6061 | 0.418  | 0.382  | 0.3675 |
| DAN-G      | 0.8335 | 0.8517 | 0.5986 | 0.4447 | 0.381  | 0.3444 |

|            |        |        |        |        |        |        |
|------------|--------|--------|--------|--------|--------|--------|
| EW-16      | 0.8154 | 0.858  | 0.7531 | 0.481  | 0.4349 | 0.3934 |
| NB7        | 0.8008 | 0.8724 | 0.7725 | 0.5737 | 0.488  | 0.4226 |
| ESS-1      | 0.834  | 0.8672 | 0.6901 | 0.4699 | 0.453  | 0.3743 |
| SK-PN-DW   | 0.8142 | 0.8614 | 0.7184 | 0.4144 | 0.3541 | 0.3954 |
| EBC-1      | 0.8102 | 0.8384 | 0.6818 | 0.3215 | 0.3151 | 0.4006 |
| EGI-1      | 0.8084 | 0.8276 | 0.6036 | 0.317  | 0.2894 | 0.3993 |
| OSC-20     | 0.8399 | 0.879  | 0.7622 | 0.5044 | 0.4422 | 0.3595 |
| NCI-H1299  | 0.8184 | 0.8448 | 0.6715 | 0.4228 | 0.371  | 0.3939 |
| KYSE-270   | 0.8632 | 0.8618 | 0.6771 | 0.4339 | 0.3562 | 0.312  |
| KINGS-1    | 0.7986 | 0.847  | 0.74   | 0.3783 | 0.3908 | 0.4402 |
| IM-9       | 0.7817 | 0.8548 | 0.776  | 0.4288 | 0.3409 | 0.4395 |
| LB1047-RCC | 0.83   | 0.8465 | 0.6933 | 0.3962 | 0.416  | 0.3765 |
| Hey        | 0.8596 | 0.8724 | 0.74   | 0.4771 | 0.4795 | 0.3344 |
| SK-N-FI    | 0.8269 | 0.8732 | 0.7269 | 0.5279 | 0.4553 | 0.3769 |
| RKO        | 0.8132 | 0.876  | 0.7565 | 0.5414 | 0.4763 | 0.3951 |
| ESO26      | 0.8189 | 0.869  | 0.7471 | 0.5069 | 0.4553 | 0.4077 |
| SU-DHL-5   | 0.8091 | 0.8339 | 0.6877 | 0.4992 | 0.4383 | 0.4191 |
| DK-MG      | 0.8369 | 0.8549 | 0.6978 | 0.4619 | 0.4093 | 0.3725 |
| HSC-4      | 0.8299 | 0.8852 | 0.7576 | 0.3989 | 0.4065 | 0.3707 |
| HCC38      | 0.8195 | 0.8878 | 0.7986 | 0.4839 | 0.4259 | 0.3998 |
| EFO-21     | 0.8073 | 0.7871 | 0.6261 | 0.3128 | 0.3032 | 0.4453 |
| BICR10     | 0.8171 | 0.8539 | 0.6975 | 0.4818 | 0.4149 | 0.3969 |
| PA-TU-8902 | 0.8312 | 0.8314 | 0.6697 | 0.3336 | 0.3349 | 0.3739 |
| DBTRG-05MG | 0.826  | 0.8664 | 0.7459 | 0.4065 | 0.3784 | 0.3786 |
| KATOIII    | 0.8446 | 0.887  | 0.8085 | 0.4578 | 0.3628 | 0.3503 |
| HCT-116    | 0.8038 | 0.8793 | 0.7687 | 0.4124 | 0.3794 | 0.3987 |
| LU-99A     | 0.8229 | 0.8915 | 0.8036 | 0.4747 | 0.4704 | 0.3867 |
| Detroit562 | 0.8003 | 0.8608 | 0.7922 | 0.4112 | 0.4157 | 0.4226 |
| SK-MEL-2   | 0.8329 | 0.8657 | 0.7333 | 0.4498 | 0.4339 | 0.3696 |
| OVISE      | 0.8218 | 0.8653 | 0.7055 | 0.4784 | 0.4291 | 0.3856 |
| MDA-MB-231 | 0.7992 | 0.8836 | 0.7518 | 0.4957 | 0.3806 | 0.4219 |
| EPLC-272H  | 0.8428 | 0.8763 | 0.7222 | 0.4479 | 0.4027 | 0.3443 |
| KYSE-410   | 0.7857 | 0.8551 | 0.7811 | 0.4282 | 0.4177 | 0.4502 |
| HT55       | 0.8475 | 0.8301 | 0.6289 | 0.2938 | 0.3177 | 0.347  |
| KLE        | 0.812  | 0.8438 | 0.6596 | 0.3555 | 0.2774 | 0.3973 |
| LS-411N    | 0.7893 | 0.8264 | 0.6555 | 0.3156 | 0.3012 | 0.4491 |
| NCI-H1155  | 0.8295 | 0.8841 | 0.7517 | 0.5225 | 0.4333 | 0.3758 |
| OSC-19     | 0.8423 | 0.8618 | 0.7434 | 0.3703 | 0.3827 | 0.3551 |
| T98G       | 0.8359 | 0.8703 | 0.7246 | 0.4772 | 0.4108 | 0.3617 |
| OV-90      | 0.8295 | 0.8696 | 0.6948 | 0.4342 | 0.4221 | 0.3796 |
| PANC-02-03 | 0.8132 | 0.8061 | 0.6    | 0.3388 | 0.3354 | 0.4066 |
| LXF-289    | 0.7858 | 0.8621 | 0.7771 | 0.4873 | 0.4121 | 0.4261 |
| TE-4       | 0.8325 | 0.8542 | 0.7158 | 0.4506 | 0.4258 | 0.3697 |

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|--------------|--------|--------|--------|--------|--------|--------|
| NCI-H1869    | 0.7905 | 0.8395 | 0.7152 | 0.4624 | 0.375  | 0.4197 |
| OVK-18       | 0.7985 | 0.8724 | 0.7972 | 0.4474 | 0.3972 | 0.4057 |
| KYAE-1       | 0.8522 | 0.8599 | 0.6813 | 0.4038 | 0.328  | 0.3319 |
| NCI-H520     | 0.8021 | 0.86   | 0.7433 | 0.4886 | 0.4498 | 0.4358 |
| MDST8        | 0.8102 | 0.8535 | 0.6629 | 0.3383 | 0.2658 | 0.3928 |
| KMS-11       | 0.8207 | 0.8745 | 0.7712 | 0.4404 | 0.4338 | 0.4093 |
| TE-9         | 0.8477 | 0.8757 | 0.7403 | 0.4831 | 0.4203 | 0.3472 |
| MHH-ES-1     | 0.8021 | 0.8507 | 0.7546 | 0.4839 | 0.4198 | 0.4026 |
| SCC-9        | 0.8427 | 0.8703 | 0.6901 | 0.4477 | 0.4206 | 0.3417 |
| A375         | 0.8172 | 0.872  | 0.7388 | 0.4877 | 0.385  | 0.3754 |
| CAL-51       | 0.8132 | 0.8912 | 0.8344 | 0.5039 | 0.4587 | 0.4014 |
| NCI-H3118    | 0.8091 | 0.8555 | 0.7134 | 0.4571 | 0.4058 | 0.4136 |
| LoVo         | 0.7986 | 0.8797 | 0.7724 | 0.4415 | 0.393  | 0.4105 |
| COLO-680N    | 0.8231 | 0.8449 | 0.7254 | 0.4429 | 0.4039 | 0.376  |
| MDA-MB-361   | 0.7806 | 0.8433 | 0.7222 | 0.4019 | 0.3556 | 0.4543 |
| TOV-112D     | 0.8197 | 0.8989 | 0.8218 | 0.5515 | 0.4803 | 0.3901 |
| MFE-296      | 0.8003 | 0.8483 | 0.6934 | 0.3925 | 0.3565 | 0.4212 |
| OVCAR-5      | 0.8073 | 0.8549 | 0.7684 | 0.4812 | 0.4511 | 0.434  |
| NCI-H1944    | 0.8364 | 0.8852 | 0.7887 | 0.4521 | 0.4436 | 0.3577 |
| MCF7         | 0.8388 | 0.8118 | 0.5405 | 0.1597 | 0.2241 | 0.3459 |
| LU-65        | 0.8358 | 0.8321 | 0.6549 | 0.3953 | 0.3442 | 0.3696 |
| RERF-LC-Sq1  | 0.8171 | 0.8417 | 0.6974 | 0.4357 | 0.403  | 0.3953 |
| RL95-2       | 0.8532 | 0.8883 | 0.7467 | 0.5535 | 0.4706 | 0.3152 |
| RMG-I        | 0.8165 | 0.8719 | 0.7517 | 0.4841 | 0.409  | 0.4028 |
| HSC-39       | 0.8306 | 0.856  | 0.6933 | 0.396  | 0.4371 | 0.3863 |
| TMK-1        | 0.8012 | 0.8474 | 0.7389 | 0.4512 | 0.3802 | 0.4218 |
| OVCAR-8      | 0.8038 | 0.8873 | 0.7692 | 0.5787 | 0.471  | 0.4033 |
| ES-2         | 0.8206 | 0.8423 | 0.6737 | 0.4917 | 0.4539 | 0.3997 |
| A549         | 0.8262 | 0.826  | 0.6613 | 0.3546 | 0.3294 | 0.3731 |
| PE-CA-PJ15   | 0.813  | 0.8564 | 0.7341 | 0.4595 | 0.4417 | 0.3965 |
| NB17         | 0.8113 | 0.8691 | 0.7761 | 0.4297 | 0.391  | 0.3973 |
| D-542MG      | 0.8457 | 0.8682 | 0.7407 | 0.4654 | 0.4301 | 0.3567 |
| HuP-T4       | 0.8295 | 0.8607 | 0.7182 | 0.3573 | 0.3503 | 0.3691 |
| TOV-21G      | 0.8285 | 0.8825 | 0.752  | 0.4228 | 0.3705 | 0.3533 |
| AM-38        | 0.8159 | 0.8721 | 0.7981 | 0.4958 | 0.43   | 0.4026 |
| MFM-223      | 0.7787 | 0.8298 | 0.7    | 0.4268 | 0.3851 | 0.4301 |
| YH-13        | 0.9231 | 0.8951 | 0.6864 | 0.43   | 0.2956 | 0.2032 |
| SF539        | 0.8166 | 0.8344 | 0.6933 | 0.4284 | 0.3977 | 0.3939 |
| UACC-893     | 0.8143 | 0.851  | 0.7627 | 0.3438 | 0.3607 | 0.3932 |
| KNS-62       | 0.8295 | 0.8522 | 0.6711 | 0.4157 | 0.4057 | 0.3867 |
| COLO-320-HSR | 0.8184 | 0.8634 | 0.7407 | 0.4518 | 0.4348 | 0.3849 |
| SNU-C5       | 0.833  | 0.8783 | 0.7646 | 0.4943 | 0.4277 | 0.371  |
| CHP-212      | 0.851  | 0.8816 | 0.6774 | 0.4324 | 0.3962 | 0.3365 |

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|------------|--------|--------|--------|--------|--------|--------|
| OAW-42     | 0.7816 | 0.7949 | 0.6796 | 0.2789 | 0.2718 | 0.4645 |
| PANC-03-27 | 0.841  | 0.8528 | 0.7092 | 0.4174 | 0.4228 | 0.3605 |
| A2058      | 0.8018 | 0.8271 | 0.6943 | 0.4365 | 0.4182 | 0.4273 |
| EW-22      | 0.8027 | 0.8763 | 0.8028 | 0.4499 | 0.4021 | 0.3993 |
| HGC-27     | 0.7927 | 0.87   | 0.7749 | 0.4843 | 0.454  | 0.4277 |
| LK-2       | 0.7998 | 0.8673 | 0.7867 | 0.4248 | 0.4069 | 0.4408 |
| ROS-50     | 0.8056 | 0.8346 | 0.7126 | 0.3273 | 0.2707 | 0.4102 |
| RPMI-8226  | 0.8254 | 0.8648 | 0.7304 | 0.3869 | 0.359  | 0.3734 |
| CL-11      | 0.8369 | 0.8743 | 0.7164 | 0.4703 | 0.4068 | 0.3523 |
| C2BBe1     | 0.8172 | 0.8534 | 0.7109 | 0.3563 | 0.3669 | 0.3814 |
| NCI-H650   | 0.8544 | 0.8711 | 0.7889 | 0.4363 | 0.3622 | 0.3368 |
| MDA-MB-453 | 0.8125 | 0.7783 | 0.5393 | 0.1844 | 0.2297 | 0.4115 |
| SNU-1      | 0.8358 | 0.8482 | 0.6616 | 0.4198 | 0.3797 | 0.3669 |
| 769-P      | 0.8349 | 0.8474 | 0.6718 | 0.4337 | 0.4239 | 0.3872 |
| HCT-15     | 0.8381 | 0.8543 | 0.6964 | 0.4239 | 0.457  | 0.3803 |
| OVCAR-3    | 0.8434 | 0.8536 | 0.6466 | 0.4208 | 0.39   | 0.3468 |
| KYSE-150   | 0.8165 | 0.886  | 0.7708 | 0.5385 | 0.4134 | 0.3747 |
| NCI-N87    | 0.7934 | 0.8358 | 0.6993 | 0.3928 | 0.3604 | 0.4274 |
| SF295      | 0.8324 | 0.8711 | 0.7749 | 0.5344 | 0.5068 | 0.3935 |
| CAPAN-1    | 0.8003 | 0.8529 | 0.7    | 0.41   | 0.3797 | 0.4096 |
| FADU       | 0.7909 | 0.8763 | 0.7987 | 0.5177 | 0.4144 | 0.4278 |
| TYK-nu     | 0.8097 | 0.8791 | 0.7692 | 0.5312 | 0.445  | 0.4038 |
| SNU-61     | 0.7811 | 0.8696 | 0.7674 | 0.4435 | 0.355  | 0.4286 |
| HuP-T3     | 0.8364 | 0.8652 | 0.7083 | 0.4458 | 0.3769 | 0.3484 |
| D-502MG    | 0.8522 | 0.8449 | 0.6111 | 0.3104 | 0.3022 | 0.3327 |
| SCC90      | 0.8561 | 0.8965 | 0.7143 | 0.4238 | 0.3778 | 0.325  |
| LCLC-97TM1 | 0.8526 | 0.872  | 0.7342 | 0.4172 | 0.3144 | 0.3312 |
| T84        | 0.8632 | 0.8894 | 0.6737 | 0.4384 | 0.249  | 0.2835 |
| HCC1395    | 0.8423 | 0.8275 | 0.65   | 0.3743 | 0.3242 | 0.3479 |
| LS-513     | 0.8142 | 0.8607 | 0.713  | 0.3819 | 0.3395 | 0.3907 |
| NH-12      | 0.7752 | 0.8527 | 0.7481 | 0.3981 | 0.3435 | 0.4254 |
| SNB75      | 0.8542 | 0.8728 | 0.7197 | 0.3961 | 0.3645 | 0.3333 |
| SW626      | 0.793  | 0.8277 | 0.6588 | 0.449  | 0.3751 | 0.4162 |
| OVMIU      | 0.7991 | 0.7564 | 0.5785 | 0.2867 | 0.2887 | 0.444  |
| OACM5-1    | 0.8334 | 0.8465 | 0.709  | 0.4202 | 0.3992 | 0.3678 |
| HCC1806    | 0.8143 | 0.8545 | 0.7133 | 0.4242 | 0.4015 | 0.4041 |
| JHU-011    | 0.8417 | 0.8698 | 0.6691 | 0.4019 | 0.4061 | 0.3593 |
| NCI-H1650  | 0.8369 | 0.8604 | 0.754  | 0.4742 | 0.4043 | 0.3694 |
| HCC1143    | 0.8207 | 0.8584 | 0.7244 | 0.4106 | 0.374  | 0.3988 |
| LNZTA3WT4  | 0.8096 | 0.8279 | 0.6524 | 0.418  | 0.3956 | 0.4085 |
| NCI-H1568  | 0.8341 | 0.852  | 0.6977 | 0.4319 | 0.3871 | 0.3971 |
| KYSE-220   | 0.8166 | 0.858  | 0.7444 | 0.5288 | 0.4597 | 0.402  |
| NCI-H1975  | 0.8288 | 0.8507 | 0.7191 | 0.5019 | 0.4655 | 0.3999 |

|                 |        |        |        |        |        |        |
|-----------------|--------|--------|--------|--------|--------|--------|
| DiFi            | 0.8078 | 0.8339 | 0.6719 | 0.349  | 0.3426 | 0.4108 |
| HCC2998         | 0.8166 | 0.8609 | 0.719  | 0.4012 | 0.3558 | 0.3945 |
| OCI-LY-19       | 0.8707 | 0.8248 | 0.5116 | 0.276  | 0.2839 | 0.2923 |
| PL4             | 0.8212 | 0.8353 | 0.708  | 0.4276 | 0.3872 | 0.3928 |
| LNCaP-Clone-FGC | 0.8201 | 0.8595 | 0.7383 | 0.3255 | 0.3383 | 0.3885 |
| OCI-AML2        | 0.8225 | 0.8798 | 0.7267 | 0.4313 | 0.4168 | 0.3762 |
| T-T             | 0.8855 | 0.8574 | 0.5342 | 0.3273 | 0.2621 | 0.2797 |
| HARA            | 0.8446 | 0.8666 | 0.7073 | 0.4757 | 0.4649 | 0.3574 |
| PC-14           | 0.8184 | 0.8537 | 0.7261 | 0.4588 | 0.4222 | 0.3989 |
| A2780ADR        | 0.7838 | 0.8647 | 0.7809 | 0.4989 | 0.433  | 0.4517 |
| PSN1            | 0.8161 | 0.855  | 0.7063 | 0.4642 | 0.417  | 0.3817 |
| no-10           | 0.8207 | 0.8699 | 0.7153 | 0.4981 | 0.4008 | 0.3819 |
| MDA-MB-415      | 0.8352 | 0.8645 | 0.6937 | 0.4396 | 0.3524 | 0.3404 |
| LS-180          | 0.8551 | 0.8875 | 0.735  | 0.4069 | 0.4086 | 0.3134 |
| SU8686          | 0.8247 | 0.8423 | 0.6881 | 0.3917 | 0.3395 | 0.385  |
| PCI-15A         | 0.8347 | 0.8694 | 0.7516 | 0.4248 | 0.4475 | 0.3835 |
| Hs-683          | 0.82   | 0.8435 | 0.774  | 0.4428 | 0.4224 | 0.401  |
| MDA-MB-468      | 0.8248 | 0.866  | 0.741  | 0.4505 | 0.4063 | 0.3715 |
| SW1088          | 0.7887 | 0.8428 | 0.7283 | 0.4624 | 0.4247 | 0.4457 |
| COLO-824        | 0.8515 | 0.8966 | 0.75   | 0.5015 | 0.3609 | 0.3171 |
| NCI-H1755       | 0.82   | 0.861  | 0.753  | 0.5022 | 0.4472 | 0.3997 |
| IM-95           | 0.8388 | 0.852  | 0.6778 | 0.3843 | 0.3058 | 0.3496 |
| Ca9-22          | 0.8393 | 0.8958 | 0.7719 | 0.5417 | 0.4889 | 0.336  |
| KYSE-70         | 0.8196 | 0.8597 | 0.7222 | 0.4925 | 0.4015 | 0.3751 |
| COLO-205        | 0.8346 | 0.8769 | 0.7236 | 0.4173 | 0.3853 | 0.3683 |
| KP-N-YS         | 0.8009 | 0.8448 | 0.6842 | 0.389  | 0.3782 | 0.4178 |
| EW-7            | 0.8463 | 0.8775 | 0.6914 | 0.4667 | 0.4783 | 0.3414 |
| BxPC-3          | 0.8353 | 0.8592 | 0.6449 | 0.3852 | 0.3278 | 0.3421 |
| KP-4            | 0.8276 | 0.8802 | 0.7178 | 0.52   | 0.4415 | 0.3505 |
| JHOS-4          | 0.8108 | 0.8623 | 0.7794 | 0.3766 | 0.3948 | 0.4072 |
| CAL-33          | 0.7986 | 0.8732 | 0.8    | 0.3789 | 0.3755 | 0.4047 |
| OC-314          | 0.7997 | 0.8445 | 0.7748 | 0.4044 | 0.4048 | 0.4511 |
| RCC-FG2         | 0.8323 | 0.8507 | 0.7071 | 0.421  | 0.4074 | 0.384  |
| PCI-38          | 0.8061 | 0.8699 | 0.7833 | 0.4884 | 0.4585 | 0.4293 |
| A2780           | 0.789  | 0.871  | 0.8051 | 0.4692 | 0.4023 | 0.4195 |
| HT-29           | 0.802  | 0.8444 | 0.7097 | 0.4339 | 0.4371 | 0.432  |

**Supplementary Table 5. Comparisons of performance of PIC with existing sequence-based methods.** EP-EDL and EP-GBDT are both cell line non-specific models based, while DeepCellEss is a cell line-specific models trained on 323 human cell lines. In the DeepCellEss paper, only AUROC and AUPRC were calculated. The comparison between PIC and DeepCellEss was the median values of AUROC and AUPRC on 323 human cell lines. The best performance values for each metric were highlighted in bold.

| Model       | Accuracy      | Recall        | Precision     | F1            | AUROC         | AUPRC         |
|-------------|---------------|---------------|---------------|---------------|---------------|---------------|
| DeepCellEss | NA            | NA            | NA            | NA            | 0.7570        | 0.7532        |
| PIC         | NA            | NA            | NA            | NA            | <b>0.8534</b> | <b>0.8584</b> |
| EP-EDL      | 0.8159        | <b>0.8730</b> | 0.7474        | 0.8258        | 0.8951        | 0.8740        |
| PIC         | <b>0.8587</b> | 0.8667        | <b>0.8531</b> | <b>0.8598</b> | <b>0.9375</b> | <b>0.9286</b> |
| EP-GBDT     | 0.7300        | 0.7870        | 0.8110        | 0.7450        | 0.8150        | 0.8110        |
| PIC         | <b>0.8520</b> | <b>0.8540</b> | <b>0.8510</b> | <b>0.8530</b> | <b>0.9360</b> | <b>0.9280</b> |
| PIC-BASE    | 0.7234        | <b>0.6809</b> | 0.7442        | 0.7111        | 0.7944        | 0.7697        |
| PIC         | <b>0.7570</b> | 0.6773        | <b>0.8059</b> | <b>0.7360</b> | <b>0.8457</b> | <b>0.8190</b> |

**Supplementary Table 6. Detailed performance comparison results between PIC and DeepCellEss.**

| Cell line  | PIC    |        | DeepCellEss |       |
|------------|--------|--------|-------------|-------|
|            | AUROC  | AUPRC  | AUROC       | AUPRC |
| RERF-GC-1B | 0.8219 | 0.8387 | 0.72        | 0.723 |
| A172       | 0.859  | 0.8573 | 0.756       | 0.727 |
| KP-1N      | 0.8674 | 0.8751 | 0.76        | 0.752 |
| D-423MG    | 0.8185 | 0.8277 | 0.713       | 0.728 |
| HCE-4      | 0.8614 | 0.8615 | 0.774       | 0.775 |
| MOG-G-UVW  | 0.8691 | 0.8838 | 0.759       | 0.751 |
| IGROV-1    | 0.8135 | 0.8313 | 0.739       | 0.729 |
| SCC-4      | 0.8455 | 0.8514 | 0.753       | 0.75  |
| HCC-15     | 0.8116 | 0.8165 | 0.769       | 0.76  |
| LB771-HNC  | 0.8688 | 0.8741 | 0.8         | 0.807 |
| Caov-4     | 0.8464 | 0.8503 | 0.746       | 0.74  |
| SF126      | 0.8536 | 0.8569 | 0.78        | 0.784 |
| PCI-30     | 0.8411 | 0.8488 | 0.731       | 0.719 |
| NCI-H226   | 0.8293 | 0.838  | 0.729       | 0.742 |
| 22RV1      | 0.8472 | 0.8597 | 0.749       | 0.747 |

|             |        |        |       |       |
|-------------|--------|--------|-------|-------|
| PA-TU-8988T | 0.875  | 0.8801 | 0.763 | 0.766 |
| HSC-3       | 0.8759 | 0.871  | 0.786 | 0.78  |
| SK-N-SH     | 0.8388 | 0.8375 | 0.73  | 0.713 |
| KYSE-450    | 0.8465 | 0.8469 | 0.742 | 0.721 |
| SW1573      | 0.8889 | 0.8956 | 0.781 | 0.771 |
| BB30-HNC    | 0.8794 | 0.8885 | 0.802 | 0.809 |
| 23132-87    | 0.8526 | 0.8558 | 0.767 | 0.752 |
| OE33        | 0.8602 | 0.86   | 0.784 | 0.781 |
| MKN28       | 0.8622 | 0.8563 | 0.734 | 0.72  |
| UWB1.289    | 0.847  | 0.849  | 0.757 | 0.758 |
| KP-N-YN     | 0.8741 | 0.8834 | 0.797 | 0.778 |
| EVSA-T      | 0.8738 | 0.8805 | 0.778 | 0.769 |
| SNU-16      | 0.859  | 0.8566 | 0.77  | 0.767 |
| SK-N-AS     | 0.8297 | 0.8414 | 0.755 | 0.749 |
| Becker      | 0.8557 | 0.8635 | 0.756 | 0.749 |
| EC-GI-10    | 0.8507 | 0.8667 | 0.737 | 0.728 |
| SNU-81      | 0.8551 | 0.8633 | 0.745 | 0.735 |
| OE21        | 0.881  | 0.8841 | 0.767 | 0.76  |
| GAMG        | 0.831  | 0.8456 | 0.71  | 0.714 |
| EFO-27      | 0.8434 | 0.8388 | 0.752 | 0.747 |
| RKN         | 0.8027 | 0.8081 | 0.731 | 0.728 |
| CAL-27      | 0.8564 | 0.8669 | 0.766 | 0.768 |
| KYSE-520    | 0.8696 | 0.8617 | 0.768 | 0.759 |
| HCC1937     | 0.8369 | 0.8361 | 0.715 | 0.695 |
| COLO-684    | 0.8471 | 0.8551 | 0.781 | 0.792 |
| NB10        | 0.8497 | 0.8609 | 0.742 | 0.73  |
| A2780cis    | 0.8688 | 0.8547 | 0.77  | 0.772 |
| NCI-H1355   | 0.8549 | 0.86   | 0.78  | 0.767 |
| NCI-H322M   | 0.81   | 0.8325 | 0.761 | 0.764 |
| HO-1-u-1    | 0.8645 | 0.8661 | 0.741 | 0.739 |
| NCI-H3122   | 0.8225 | 0.8395 | 0.742 | 0.739 |
| SW620       | 0.8756 | 0.887  | 0.763 | 0.751 |
| NCI-H2170   | 0.8571 | 0.8683 | 0.756 | 0.741 |
| 8-MG-BA     | 0.8662 | 0.8845 | 0.777 | 0.782 |
| SK-MG-1     | 0.8419 | 0.8384 | 0.754 | 0.739 |
| DOV13       | 0.847  | 0.8442 | 0.745 | 0.718 |
| SKN-3       | 0.8808 | 0.8833 | 0.758 | 0.746 |
| U-87-MG     | 0.8364 | 0.8518 | 0.752 | 0.749 |
| SAS         | 0.8946 | 0.8943 | 0.766 | 0.765 |
| DOK         | 0.8845 | 0.8915 | 0.78  | 0.787 |
| NB6         | 0.8411 | 0.8504 | 0.758 | 0.761 |
| IST-MEL1    | 0.8282 | 0.8383 | 0.738 | 0.752 |
| T47D        | 0.8767 | 0.8786 | 0.788 | 0.791 |

|             |        |        |       |       |
|-------------|--------|--------|-------|-------|
| PCI-6A      | 0.8874 | 0.8867 | 0.764 | 0.756 |
| LS-123      | 0.8763 | 0.8857 | 0.79  | 0.809 |
| GI-1        | 0.8331 | 0.8402 | 0.758 | 0.749 |
| NCI-H2023   | 0.8626 | 0.876  | 0.777 | 0.761 |
| SW837       | 0.8746 | 0.8807 | 0.779 | 0.773 |
| OCUB-M      | 0.8784 | 0.8827 | 0.802 | 0.8   |
| MC-IXC      | 0.8442 | 0.8556 | 0.752 | 0.73  |
| CAL-72      | 0.8553 | 0.8623 | 0.781 | 0.773 |
| BE2-M17     | 0.8351 | 0.8461 | 0.741 | 0.734 |
| PANC-08-13  | 0.8071 | 0.8003 | 0.737 | 0.733 |
| SW1990      | 0.8614 | 0.8595 | 0.786 | 0.789 |
| TGW         | 0.8439 | 0.84   | 0.762 | 0.734 |
| PEO4        | 0.8227 | 0.8472 | 0.734 | 0.73  |
| HPAF-II     | 0.8723 | 0.88   | 0.763 | 0.783 |
| KM12        | 0.8664 | 0.8637 | 0.789 | 0.791 |
| HCC-78      | 0.809  | 0.7878 | 0.718 | 0.706 |
| FLO-1       | 0.8564 | 0.861  | 0.736 | 0.732 |
| BICR78      | 0.8549 | 0.842  | 0.768 | 0.766 |
| SJSA-1      | 0.8748 | 0.8864 | 0.762 | 0.765 |
| VMRC-LCD    | 0.8647 | 0.8661 | 0.756 | 0.756 |
| SK-GT-4     | 0.8726 | 0.8732 | 0.763 | 0.764 |
| KP-3        | 0.838  | 0.863  | 0.764 | 0.782 |
| SU-DHL-10   | 0.8361 | 0.8451 | 0.712 | 0.699 |
| SU-DHL-8    | 0.8604 | 0.8659 | 0.764 | 0.759 |
| OPM-2       | 0.8471 | 0.8382 | 0.761 | 0.766 |
| BICR22      | 0.8947 | 0.8947 | 0.768 | 0.765 |
| NB69        | 0.8519 | 0.8508 | 0.751 | 0.749 |
| JHU-022     | 0.8582 | 0.8653 | 0.778 | 0.768 |
| SCH         | 0.8387 | 0.8431 | 0.749 | 0.74  |
| hTERT-RPE-1 | 0.8527 | 0.8565 | 0.764 | 0.767 |
| SUIT-2      | 0.8209 | 0.8257 | 0.76  | 0.766 |
| JIMT-1      | 0.8588 | 0.8598 | 0.753 | 0.766 |
| NCI-H1993   | 0.8212 | 0.8301 | 0.732 | 0.734 |
| PANC-04-03  | 0.8315 | 0.8482 | 0.767 | 0.762 |
| LOU-NH91    | 0.8287 | 0.8477 | 0.75  | 0.755 |
| OCI-AML3    | 0.8809 | 0.8814 | 0.766 | 0.758 |
| TE-5        | 0.8573 | 0.8459 | 0.79  | 0.799 |
| COR-L23     | 0.8387 | 0.8452 | 0.769 | 0.767 |
| NB13        | 0.8265 | 0.8277 | 0.72  | 0.713 |
| GP5d        | 0.8761 | 0.8699 | 0.809 | 0.8   |
| HEC-1       | 0.8364 | 0.8395 | 0.753 | 0.756 |
| COLO-678    | 0.7976 | 0.8039 | 0.771 | 0.773 |
| no-11       | 0.833  | 0.8468 | 0.743 | 0.742 |

|            |        |        |       |       |
|------------|--------|--------|-------|-------|
| HCC1954    | 0.8708 | 0.8753 | 0.773 | 0.77  |
| 42-MG-BA   | 0.84   | 0.844  | 0.761 | 0.755 |
| GB-1       | 0.8564 | 0.863  | 0.783 | 0.777 |
| KURAMOCHI  | 0.836  | 0.8464 | 0.719 | 0.705 |
| RCM-1      | 0.8428 | 0.8488 | 0.753 | 0.75  |
| EW-1       | 0.873  | 0.8716 | 0.768 | 0.758 |
| BHY        | 0.8523 | 0.8518 | 0.761 | 0.739 |
| ESO51      | 0.8273 | 0.8318 | 0.739 | 0.726 |
| L-363      | 0.8521 | 0.8603 | 0.757 | 0.753 |
| NCI-H1915  | 0.8435 | 0.858  | 0.772 | 0.777 |
| TE-8       | 0.851  | 0.8547 | 0.743 | 0.753 |
| AU565      | 0.844  | 0.8539 | 0.718 | 0.723 |
| KNS-42     | 0.8172 | 0.8249 | 0.704 | 0.726 |
| MDA-MB-436 | 0.8717 | 0.8826 | 0.731 | 0.723 |
| LP-1       | 0.879  | 0.89   | 0.772 | 0.775 |
| NCI-H358   | 0.848  | 0.8629 | 0.745 | 0.744 |
| PCI-4B     | 0.8625 | 0.8622 | 0.765 | 0.758 |
| OVCA420    | 0.8655 | 0.8838 | 0.782 | 0.776 |
| AGS        | 0.8473 | 0.8543 | 0.751 | 0.756 |
| KYSE-140   | 0.8662 | 0.867  | 0.796 | 0.79  |
| NUGC-3     | 0.8531 | 0.847  | 0.761 | 0.755 |
| JHOS-2     | 0.8287 | 0.8414 | 0.719 | 0.698 |
| NB5        | 0.8529 | 0.8544 | 0.763 | 0.76  |
| HOP-62     | 0.8821 | 0.8825 | 0.792 | 0.781 |
| NMC-G1     | 0.8615 | 0.8663 | 0.806 | 0.797 |
| ES4        | 0.8741 | 0.8694 | 0.756 | 0.755 |
| SNU-C1     | 0.8644 | 0.8723 | 0.825 | 0.826 |
| NCI-H23    | 0.8629 | 0.864  | 0.769 | 0.78  |
| U251       | 0.8785 | 0.8712 | 0.787 | 0.789 |
| SW48       | 0.8434 | 0.8392 | 0.761 | 0.773 |
| MIA-PaCa-2 | 0.8574 | 0.8583 | 0.749 | 0.755 |
| CAS-1      | 0.8552 | 0.8561 | 0.765 | 0.762 |
| NCI-H2087  | 0.8265 | 0.8433 | 0.702 | 0.718 |
| PEO1       | 0.8225 | 0.8315 | 0.716 | 0.711 |
| Hs-578-T   | 0.8538 | 0.8604 | 0.741 | 0.734 |
| HT-29      | 0.8423 | 0.8535 | 0.746 | 0.745 |
| SK-N-DZ    | 0.8428 | 0.8434 | 0.737 | 0.742 |
| ES5        | 0.8442 | 0.8469 | 0.737 | 0.748 |
| TE-10      | 0.8817 | 0.8831 | 0.784 | 0.769 |
| AsPC-1     | 0.8431 | 0.8507 | 0.73  | 0.725 |
| PANC-10-05 | 0.8587 | 0.8635 | 0.748 | 0.759 |
| SNG-M      | 0.8565 | 0.8706 | 0.731 | 0.726 |
| JHU-029    | 0.8473 | 0.8663 | 0.767 | 0.77  |

|            |        |        |       |       |
|------------|--------|--------|-------|-------|
| LC-1-sq    | 0.8436 | 0.8367 | 0.738 | 0.741 |
| HCC70      | 0.8731 | 0.8816 | 0.784 | 0.791 |
| MFE-280    | 0.8468 | 0.8457 | 0.768 | 0.764 |
| DU-145     | 0.866  | 0.8673 | 0.772 | 0.765 |
| TC-71      | 0.862  | 0.8732 | 0.764 | 0.76  |
| LN-229     | 0.8612 | 0.8567 | 0.765 | 0.751 |
| A253       | 0.8451 | 0.8411 | 0.744 | 0.731 |
| KYSE-510   | 0.8534 | 0.8638 | 0.751 | 0.744 |
| ES8        | 0.8818 | 0.8767 | 0.767 | 0.766 |
| SK-MES-1   | 0.8365 | 0.8498 | 0.753 | 0.754 |
| KON        | 0.8764 | 0.8758 | 0.755 | 0.75  |
| HCC1187    | 0.8671 | 0.8722 | 0.76  | 0.76  |
| ARH-77     | 0.8559 | 0.8454 | 0.754 | 0.73  |
| DAN-G      | 0.8553 | 0.8572 | 0.746 | 0.74  |
| EW-16      | 0.8473 | 0.8525 | 0.746 | 0.748 |
| NB7        | 0.8648 | 0.8694 | 0.75  | 0.737 |
| ESS-1      | 0.852  | 0.856  | 0.749 | 0.741 |
| SK-PN-DW   | 0.8713 | 0.8693 | 0.793 | 0.783 |
| EBC-1      | 0.848  | 0.8522 | 0.739 | 0.746 |
| EGI-1      | 0.8532 | 0.8524 | 0.795 | 0.803 |
| OSC-20     | 0.8594 | 0.8735 | 0.796 | 0.791 |
| NCI-H1299  | 0.8829 | 0.8842 | 0.776 | 0.778 |
| KYSE-270   | 0.868  | 0.8801 | 0.74  | 0.744 |
| KINGS-1    | 0.8445 | 0.8429 | 0.763 | 0.761 |
| IM-9       | 0.8533 | 0.8425 | 0.732 | 0.727 |
| LB1047-RCC | 0.857  | 0.8536 | 0.766 | 0.753 |
| Hey        | 0.8299 | 0.8424 | 0.745 | 0.747 |
| SK-N-FI    | 0.8429 | 0.8521 | 0.732 | 0.712 |
| RKO        | 0.8582 | 0.8716 | 0.783 | 0.764 |
| ESO26      | 0.849  | 0.8566 | 0.735 | 0.72  |
| SU-DHL-5   | 0.8291 | 0.8377 | 0.771 | 0.769 |
| DK-MG      | 0.8525 | 0.867  | 0.722 | 0.714 |
| HSC-4      | 0.8813 | 0.8817 | 0.784 | 0.772 |
| HCC38      | 0.8602 | 0.8576 | 0.764 | 0.757 |
| EFO-21     | 0.7989 | 0.8286 | 0.7   | 0.703 |
| BICR10     | 0.8636 | 0.8744 | 0.766 | 0.772 |
| PA-TU-8902 | 0.8431 | 0.8368 | 0.762 | 0.77  |
| DBTRG-05MG | 0.8506 | 0.8481 | 0.74  | 0.732 |
| KATOIII    | 0.8639 | 0.8765 | 0.75  | 0.763 |
| HCT-116    | 0.8587 | 0.8616 | 0.782 | 0.795 |
| LU-99A     | 0.8663 | 0.8628 | 0.758 | 0.753 |
| LS1034     | 0.8709 | 0.8607 | 0.785 | 0.784 |
| Detroit562 | 0.8488 | 0.8495 | 0.744 | 0.723 |

|             |        |        |       |       |
|-------------|--------|--------|-------|-------|
| SK-MEL-2    | 0.8665 | 0.8704 | 0.806 | 0.798 |
| OVISE       | 0.8427 | 0.8428 | 0.726 | 0.725 |
| MDA-MB-231  | 0.8341 | 0.8376 | 0.731 | 0.733 |
| EPLC-272H   | 0.8712 | 0.8805 | 0.78  | 0.782 |
| KYSE-410    | 0.8576 | 0.8732 | 0.752 | 0.77  |
| HT55        | 0.8673 | 0.8743 | 0.762 | 0.752 |
| KLE         | 0.8658 | 0.8767 | 0.803 | 0.798 |
| LS-411N     | 0.8396 | 0.8421 | 0.739 | 0.736 |
| NCI-H1155   | 0.8849 | 0.8931 | 0.769 | 0.768 |
| OSC-19      | 0.8454 | 0.8654 | 0.77  | 0.768 |
| T98G        | 0.874  | 0.8732 | 0.782 | 0.76  |
| OV-90       | 0.8616 | 0.8585 | 0.751 | 0.739 |
| PANC-02-03  | 0.8793 | 0.8835 | 0.773 | 0.773 |
| LXF-289     | 0.8587 | 0.8779 | 0.783 | 0.787 |
| TE-4        | 0.8659 | 0.8685 | 0.765 | 0.755 |
| NCI-H1869   | 0.824  | 0.8418 | 0.702 | 0.699 |
| OVK-18      | 0.8708 | 0.8697 | 0.768 | 0.767 |
| KYAE-1      | 0.8961 | 0.8878 | 0.765 | 0.767 |
| NCI-H520    | 0.8356 | 0.8333 | 0.742 | 0.742 |
| MDST8       | 0.8247 | 0.8404 | 0.728 | 0.731 |
| KMS-11      | 0.8336 | 0.8266 | 0.748 | 0.725 |
| TE-9        | 0.86   | 0.8745 | 0.761 | 0.767 |
| MHH-ES-1    | 0.8554 | 0.8451 | 0.749 | 0.726 |
| SCC-9       | 0.8569 | 0.8676 | 0.759 | 0.774 |
| A375        | 0.8599 | 0.8451 | 0.787 | 0.775 |
| CAL-51      | 0.8674 | 0.8752 | 0.745 | 0.725 |
| NCI-H3118   | 0.8353 | 0.8365 | 0.742 | 0.73  |
| LoVo        | 0.8446 | 0.8542 | 0.764 | 0.774 |
| COLO-680N   | 0.8122 | 0.8242 | 0.735 | 0.73  |
| MDA-MB-361  | 0.8144 | 0.8218 | 0.718 | 0.734 |
| TOV-112D    | 0.8387 | 0.8405 | 0.752 | 0.763 |
| MFE-296     | 0.8688 | 0.8674 | 0.763 | 0.759 |
| OVCAR-5     | 0.8727 | 0.8813 | 0.782 | 0.786 |
| NCI-H1944   | 0.886  | 0.8904 | 0.744 | 0.727 |
| MCF7        | 0.8357 | 0.823  | 0.736 | 0.694 |
| LU-65       | 0.8362 | 0.8464 | 0.716 | 0.716 |
| RERF-LC-Sq1 | 0.8066 | 0.8209 | 0.731 | 0.735 |
| RL95-2      | 0.8824 | 0.8961 | 0.771 | 0.773 |
| RMG-I       | 0.7868 | 0.7991 | 0.681 | 0.695 |
| HSC-39      | 0.8589 | 0.8534 | 0.756 | 0.735 |
| TMK-1       | 0.8986 | 0.8891 | 0.776 | 0.764 |
| OVCAR-8     | 0.8557 | 0.8541 | 0.767 | 0.762 |
| ES-2        | 0.8499 | 0.8605 | 0.745 | 0.751 |

|              |        |        |       |       |
|--------------|--------|--------|-------|-------|
| A549         | 0.8571 | 0.8625 | 0.745 | 0.753 |
| PE-CA-PJ15   | 0.8661 | 0.8728 | 0.747 | 0.727 |
| NB17         | 0.8392 | 0.8449 | 0.748 | 0.75  |
| D-542MG      | 0.8332 | 0.8437 | 0.761 | 0.756 |
| HuP-T4       | 0.8428 | 0.8448 | 0.783 | 0.762 |
| TOV-21G      | 0.8623 | 0.8641 | 0.753 | 0.755 |
| AM-38        | 0.8334 | 0.8314 | 0.683 | 0.682 |
| MFM-223      | 0.859  | 0.8324 | 0.763 | 0.766 |
| YH-13        | 0.8464 | 0.8637 | 0.778 | 0.781 |
| SF539        | 0.8281 | 0.8434 | 0.749 | 0.739 |
| UACC-893     | 0.8858 | 0.8911 | 0.742 | 0.747 |
| KNS-62       | 0.8708 | 0.8747 | 0.793 | 0.793 |
| COLO-320-HSR | 0.9011 | 0.8998 | 0.77  | 0.765 |
| SNU-C5       | 0.8819 | 0.8719 | 0.807 | 0.791 |
| CHP-212      | 0.8738 | 0.8854 | 0.768 | 0.763 |
| OAW-42       | 0.8078 | 0.8204 | 0.719 | 0.72  |
| PANC-03-27   | 0.8262 | 0.8396 | 0.735 | 0.74  |
| A2058        | 0.8613 | 0.8722 | 0.743 | 0.739 |
| EW-22        | 0.8677 | 0.8781 | 0.773 | 0.771 |
| HGC-27       | 0.8366 | 0.8341 | 0.756 | 0.75  |
| LK-2         | 0.8574 | 0.8633 | 0.749 | 0.751 |
| ROS-50       | 0.8337 | 0.855  | 0.738 | 0.747 |
| RPMI-8226    | 0.8605 | 0.8679 | 0.765 | 0.762 |
| CL-11        | 0.871  | 0.8784 | 0.766 | 0.776 |
| C2BBel       | 0.8859 | 0.8884 | 0.805 | 0.803 |
| NCI-H650     | 0.817  | 0.8272 | 0.716 | 0.709 |
| MDA-MB-453   | 0.8209 | 0.8257 | 0.763 | 0.767 |
| SNU-1        | 0.8271 | 0.8326 | 0.757 | 0.741 |
| 769-P        | 0.8636 | 0.8719 | 0.757 | 0.746 |
| HCT-15       | 0.8488 | 0.8418 | 0.752 | 0.74  |
| OVCAR-3      | 0.858  | 0.8691 | 0.754 | 0.761 |
| KYSE-150     | 0.862  | 0.8646 | 0.776 | 0.779 |
| NCI-N87      | 0.8663 | 0.8492 | 0.796 | 0.776 |
| SF295        | 0.8552 | 0.8587 | 0.727 | 0.721 |
| CAPAN-1      | 0.851  | 0.8526 | 0.759 | 0.751 |
| FADU         | 0.897  | 0.8979 | 0.789 | 0.777 |
| TYK-nu       | 0.8678 | 0.8705 | 0.744 | 0.739 |
| SNU-61       | 0.8495 | 0.8587 | 0.784 | 0.777 |
| HuP-T3       | 0.8739 | 0.8721 | 0.769 | 0.758 |
| D-502MG      | 0.8651 | 0.8789 | 0.785 | 0.782 |
| SCC90        | 0.8205 | 0.8413 | 0.743 | 0.748 |
| LCLC-97TM1   | 0.8288 | 0.8524 | 0.764 | 0.769 |
| T84          | 0.8499 | 0.8662 | 0.799 | 0.792 |

|                 |        |        |       |       |
|-----------------|--------|--------|-------|-------|
| HCC1395         | 0.8565 | 0.872  | 0.754 | 0.765 |
| LS-513          | 0.876  | 0.8876 | 0.747 | 0.744 |
| NH-12           | 0.849  | 0.8517 | 0.737 | 0.718 |
| SNB75           | 0.8493 | 0.8692 | 0.749 | 0.761 |
| SW626           | 0.8461 | 0.8484 | 0.735 | 0.712 |
| OVMIU           | 0.7954 | 0.8158 | 0.708 | 0.705 |
| OACM5-1         | 0.8268 | 0.8459 | 0.732 | 0.732 |
| HCC1806         | 0.8523 | 0.8566 | 0.756 | 0.756 |
| JHU-011         | 0.8707 | 0.8736 | 0.769 | 0.754 |
| NCI-H1650       | 0.8654 | 0.8801 | 0.771 | 0.757 |
| HCC1143         | 0.8654 | 0.875  | 0.759 | 0.747 |
| LNZTA3WT4       | 0.8695 | 0.8732 | 0.758 | 0.737 |
| NCI-H1568       | 0.8023 | 0.801  | 0.738 | 0.748 |
| KYSE-220        | 0.8715 | 0.8858 | 0.773 | 0.775 |
| NCI-H1975       | 0.8176 | 0.8231 | 0.731 | 0.752 |
| DiFi            | 0.8664 | 0.8673 | 0.731 | 0.734 |
| HCC2998         | 0.8737 | 0.8549 | 0.76  | 0.718 |
| OCI-LY-19       | 0.8735 | 0.8745 | 0.776 | 0.774 |
| PL4             | 0.8654 | 0.8549 | 0.741 | 0.734 |
| LNCaP-Clone-FGC | 0.8716 | 0.8606 | 0.768 | 0.741 |
| OCI-AML2        | 0.8683 | 0.8673 | 0.766 | 0.758 |
| T-T             | 0.8431 | 0.8499 | 0.781 | 0.786 |
| HARA            | 0.8478 | 0.8671 | 0.769 | 0.767 |
| PC-14           | 0.8454 | 0.8561 | 0.744 | 0.738 |
| A2780ADR        | 0.8222 | 0.8323 | 0.752 | 0.746 |
| PSN1            | 0.8856 | 0.8924 | 0.79  | 0.791 |
| no-10           | 0.8595 | 0.8666 | 0.747 | 0.744 |
| MDA-MB-415      | 0.8876 | 0.8938 | 0.771 | 0.765 |
| LS-180          | 0.8788 | 0.8818 | 0.746 | 0.749 |
| SU8686          | 0.8614 | 0.8782 | 0.774 | 0.783 |
| PCI-15A         | 0.8798 | 0.8927 | 0.773 | 0.783 |
| Hs-683          | 0.8432 | 0.8463 | 0.727 | 0.73  |
| MDA-MB-468      | 0.8328 | 0.8442 | 0.715 | 0.706 |
| SW1088          | 0.836  | 0.8293 | 0.699 | 0.713 |
| COLO-824        | 0.8947 | 0.9037 | 0.786 | 0.787 |
| NCI-H1755       | 0.8735 | 0.8718 | 0.777 | 0.756 |
| IM-95           | 0.8601 | 0.8559 | 0.767 | 0.769 |
| Ca9-22          | 0.8845 | 0.889  | 0.744 | 0.732 |
| KYSE-70         | 0.8683 | 0.8759 | 0.775 | 0.766 |
| COLO-205        | 0.831  | 0.8363 | 0.757 | 0.744 |
| KP-N-YS         | 0.8453 | 0.8449 | 0.753 | 0.724 |
| EW-7            | 0.8645 | 0.8805 | 0.764 | 0.752 |
| BxPC-3          | 0.841  | 0.8439 | 0.742 | 0.743 |

|         |        |        |       |       |
|---------|--------|--------|-------|-------|
| KP-4    | 0.8651 | 0.8682 | 0.748 | 0.737 |
| JHOS-4  | 0.8618 | 0.865  | 0.755 | 0.752 |
| CAL-33  | 0.8671 | 0.8485 | 0.77  | 0.749 |
| OC-314  | 0.8129 | 0.8154 | 0.73  | 0.717 |
| RCC-FG2 | 0.8494 | 0.857  | 0.768 | 0.756 |
| PCI-38  | 0.8682 | 0.8709 | 0.765 | 0.76  |
| A2780   | 0.8535 | 0.8495 | 0.752 | 0.751 |

**Supplementary Table 7. Detailed correlation analysis results between cPES and hPES.**

| Cell line  | Spearman correlation coefficient | P-value  |
|------------|----------------------------------|----------|
| OAW-42     | 0.3971                           | 0.00E+00 |
| NCI-H226   | 0.3646                           | 0.00E+00 |
| NCI-H650   | 0.364                            | 0.00E+00 |
| RMG-I      | 0.3448                           | 0.00E+00 |
| HSC-39     | 0.339                            | 0.00E+00 |
| NCI-H322M  | 0.3375                           | 0.00E+00 |
| EFO-21     | 0.3374                           | 0.00E+00 |
| RERF-GC-1B | 0.334                            | 0.00E+00 |
| KNS-42     | 0.3306                           | 0.00E+00 |
| NB13       | 0.3261                           | 0.00E+00 |
| ESO51      | 0.3233                           | 0.00E+00 |
| NCI-H1993  | 0.3208                           | 0.00E+00 |
| MDA-MB-361 | 0.3202                           | 0.00E+00 |
| LOU-NH91   | 0.3181                           | 0.00E+00 |
| U-87-MG    | 0.3159                           | 0.00E+00 |
| CAS-1      | 0.3157                           | 0.00E+00 |
| DK-MG      | 0.3155                           | 0.00E+00 |
| OVCAR-5    | 0.3129                           | 0.00E+00 |
| MFE-280    | 0.3109                           | 0.00E+00 |
| NCI-H3118  | 0.309                            | 0.00E+00 |
| SK-MES-1   | 0.3087                           | 0.00E+00 |
| HPAF-II    | 0.3087                           | 0.00E+00 |
| IM-95      | 0.3075                           | 0.00E+00 |
| SCC90      | 0.3075                           | 0.00E+00 |
| JHOS-2     | 0.306                            | 0.00E+00 |
| OV-90      | 0.3056                           | 0.00E+00 |
| KLE        | 0.3054                           | 0.00E+00 |
| RKN        | 0.3044                           | 0.00E+00 |
| AM-38      | 0.3032                           | 0.00E+00 |
| OVCA420    | 0.3024                           | 0.00E+00 |
| OPM-2      | 0.2995                           | 0.00E+00 |
| COLO-205   | 0.2982                           | 0.00E+00 |

|             |        |           |
|-------------|--------|-----------|
| RCC-FG2     | 0.2959 | 0.00E+00  |
| SK-N-DZ     | 0.2955 | 0.00E+00  |
| PANC-04-03  | 0.2935 | 0.00E+00  |
| NCI-H1975   | 0.2933 | 0.00E+00  |
| D-542MG     | 0.2926 | 0.00E+00  |
| SW626       | 0.2926 | 0.00E+00  |
| OSC-19      | 0.2926 | 0.00E+00  |
| SK-N-AS     | 0.2917 | 0.00E+00  |
| RERF-LC-Sq1 | 0.2902 | 0.00E+00  |
| NCI-H3122   | 0.2893 | 0.00E+00  |
| ES5         | 0.2878 | 0.00E+00  |
| OVMU        | 0.2851 | 0.00E+00  |
| IGROV-1     | 0.2848 | 0.00E+00  |
| UACC-893    | 0.2838 | 0.00E+00  |
| SCC-4       | 0.2834 | 0.00E+00  |
| KYAE-1      | 0.2813 | 0.00E+00  |
| Hey         | 0.2811 | 0.00E+00  |
| HARA        | 0.2811 | 0.00E+00  |
| NCI-H1568   | 0.2804 | 0.00E+00  |
| Hs-683      | 0.2798 | 0.00E+00  |
| SNU-16      | 0.2796 | 0.00E+00  |
| DOV13       | 0.2795 | 0.00E+00  |
| NCI-H1755   | 0.2793 | 0.00E+00  |
| TE-8        | 0.2785 | 0.00E+00  |
| PEO1        | 0.2777 | 0.00E+00  |
| SF295       | 0.2768 | 0.00E+00  |
| Caov-4      | 0.2753 | 0.00E+00  |
| NCI-H1915   | 0.2739 | 0.00E+00  |
| PANC-08-13  | 0.2734 | 0.00E+00  |
| OVISE       | 0.2719 | 0.00E+00  |
| CAL-72      | 0.2717 | 0.00E+00  |
| DOK         | 0.2707 | 0.00E+00  |
| KMS-11      | 0.2687 | 0.00E+00  |
| T-T         | 0.2681 | 0.00E+00  |
| JHU-022     | 0.2667 | 0.00E+00  |
| OE21        | 0.2664 | 0.00E+00  |
| OC-314      | 0.2653 | 0.00E+00  |
| EGI-1       | 0.2637 | 0.00E+00  |
| OCI-LY-19   | 0.2632 | 0.00E+00  |
| EC-GI-10    | 0.2631 | 0.00E+00  |
| L-363       | 0.2623 | 0.00E+00  |
| NMC-G1      | 0.2618 | 2.24E-308 |
| NCI-H1355   | 0.261  | 2.04E-306 |

|                 |        |           |
|-----------------|--------|-----------|
| A2058           | 0.2608 | 6.32E-306 |
| NB69            | 0.26   | 5.72E-304 |
| no-11           | 0.2593 | 2.46E-302 |
| SNB75           | 0.2572 | 2.77E-297 |
| SNU-C1          | 0.2572 | 1.68E-297 |
| COLO-680N       | 0.2563 | 2.53E-295 |
| OCI-AML3        | 0.2557 | 9.51E-294 |
| T98G            | 0.2552 | 1.33E-292 |
| FLO-1           | 0.2549 | 7.07E-292 |
| LNCaP-Clone-FGC | 0.2546 | 3.12E-291 |
| PANC-02-03      | 0.2535 | 1.22E-288 |
| JHOS-4          | 0.2532 | 5.00E-288 |
| OSC-20          | 0.2522 | 1.31E-285 |
| SU8686          | 0.2522 | 1.14E-285 |
| SF126           | 0.2518 | 1.34E-284 |
| NB6             | 0.2514 | 9.45E-284 |
| DiFi            | 0.2511 | 4.82E-283 |
| NCI-H520        | 0.2507 | 4.50E-282 |
| NCI-H2087       | 0.2505 | 9.66E-282 |
| SCH             | 0.2497 | 7.37E-280 |
| KINGS-1         | 0.2497 | 5.58E-280 |
| LU-99A          | 0.2495 | 2.03E-279 |
| LS-513          | 0.2495 | 1.97E-279 |
| OACM5-1         | 0.249  | 2.87E-278 |
| ROS-50          | 0.2488 | 6.61E-278 |
| BICR10          | 0.2486 | 2.66E-277 |
| PCI-15A         | 0.2485 | 3.32E-277 |
| LK-2            | 0.2483 | 1.06E-276 |
| TMK-1           | 0.2481 | 2.75E-276 |
| NCI-H1650       | 0.248  | 5.47E-276 |
| NCI-N87         | 0.248  | 5.05E-276 |
| PANC-10-05      | 0.2477 | 2.74E-275 |
| KURAMOCHI       | 0.2474 | 1.37E-274 |
| 769-P           | 0.2474 | 1.34E-274 |
| TE-4            | 0.2471 | 6.63E-274 |
| EW-16           | 0.2467 | 6.45E-273 |
| DU-145          | 0.246  | 2.34E-271 |
| SNU-1           | 0.2452 | 1.11E-269 |
| AsPC-1          | 0.2447 | 1.69E-268 |
| UWB1            | 0.2446 | 2.20E-268 |
| SK-GT-4         | 0.244  | 5.39E-267 |
| NH-12           | 0.2439 | 1.18E-266 |
| HCC1954         | 0.2439 | 1.15E-266 |

|             |        |           |
|-------------|--------|-----------|
| KNS-62      | 0.2429 | 1.78E-264 |
| IST-MEL1    | 0.2425 | 1.75E-263 |
| NB17        | 0.2422 | 5.42E-263 |
| SW1990      | 0.2417 | 9.90E-262 |
| SK-N-FI     | 0.2414 | 3.36E-261 |
| MIA-PaCa-2  | 0.2409 | 4.50E-260 |
| MCF7        | 0.2408 | 7.82E-260 |
| KM12        | 0.2407 | 1.52E-259 |
| MOG-G-UVW   | 0.2406 | 1.94E-259 |
| Hs-578-T    | 0.2403 | 8.96E-259 |
| LB1047-RCC  | 0.2396 | 3.59E-257 |
| NCI-H1155   | 0.2372 | 7.12E-252 |
| HCC38       | 0.2372 | 6.14E-252 |
| KYSE-70     | 0.2372 | 5.16E-252 |
| CAL-27      | 0.2372 | 5.01E-252 |
| PE-CA-PJ15  | 0.2365 | 2.16E-250 |
| C2BBe1      | 0.2364 | 3.25E-250 |
| EPLC-272H   | 0.2363 | 5.21E-250 |
| EVSA-T      | 0.2346 | 1.88E-246 |
| CAPAN-1     | 0.2341 | 3.37E-245 |
| TE-5        | 0.2326 | 4.52E-242 |
| A549        | 0.2326 | 3.79E-242 |
| ES-2        | 0.2324 | 1.15E-241 |
| LNZTA3WT4   | 0.2317 | 3.02E-240 |
| PEO4        | 0.231  | 1.33E-238 |
| hTERT-RPE-1 | 0.2304 | 2.13E-237 |
| ARH-77      | 0.2303 | 2.65E-237 |
| EW-22       | 0.23   | 1.26E-236 |
| JHU-011     | 0.23   | 1.24E-236 |
| LN-229      | 0.2295 | 1.76E-235 |
| SF539       | 0.2295 | 1.64E-235 |
| TGW         | 0.2295 | 1.25E-235 |
| MDA-MB-415  | 0.2286 | 1.02E-233 |
| 8-MG-BA     | 0.2284 | 2.68E-233 |
| T47D        | 0.2283 | 5.51E-233 |
| MHH-ES-1    | 0.2281 | 1.02E-232 |
| SU-DHL-10   | 0.2279 | 2.90E-232 |
| EBC-1       | 0.2278 | 5.35E-232 |
| SKN-3       | 0.2277 | 7.29E-232 |
| HCC1806     | 0.227  | 2.44E-230 |
| KP-N-YN     | 0.2265 | 2.40E-229 |
| LXF-289     | 0.2257 | 1.34E-227 |
| NCI-H1869   | 0.2251 | 1.59E-226 |

|            |        |           |
|------------|--------|-----------|
| NCI-H1299  | 0.2246 | 1.71E-225 |
| RCM-1      | 0.2243 | 6.74E-225 |
| SW1088     | 0.2238 | 8.61E-224 |
| SK-N-SH    | 0.2235 | 2.93E-223 |
| SK-MEL-2   | 0.2223 | 9.69E-221 |
| CHP-212    | 0.2219 | 5.68E-220 |
| Becker     | 0.2218 | 1.14E-219 |
| D-423MG    | 0.2211 | 1.93E-218 |
| IM-9       | 0.2205 | 4.15E-217 |
| MC-IXC     | 0.2205 | 4.08E-217 |
| VMRC-LCD   | 0.2201 | 1.98E-216 |
| SJSA-1     | 0.2199 | 6.52E-216 |
| AU565      | 0.2194 | 6.17E-215 |
| OE33       | 0.2184 | 5.39E-213 |
| PL4        | 0.2183 | 9.21E-213 |
| TYK-nu     | 0.2181 | 2.50E-212 |
| EFO-27     | 0.2181 | 2.41E-212 |
| COLO-684   | 0.2176 | 2.24E-211 |
| KATOIII    | 0.2174 | 5.39E-211 |
| OVCAR-3    | 0.2169 | 5.74E-210 |
| HCC1187    | 0.2165 | 3.15E-209 |
| PANC-03-27 | 0.2159 | 4.30E-208 |
| SUIT-2     | 0.2155 | 2.86E-207 |
| 23132-87   | 0.2153 | 8.52E-207 |
| SAS        | 0.2144 | 4.22E-205 |
| HSC-4      | 0.2144 | 3.95E-205 |
| HCC1937    | 0.2142 | 9.81E-205 |
| 22RV1      | 0.2138 | 6.04E-204 |
| COLO-678   | 0.2138 | 4.62E-204 |
| BICR22     | 0.2128 | 4.87E-202 |
| A2780ADR   | 0.2126 | 1.43E-201 |
| HEC-1      | 0.2118 | 4.02E-200 |
| KON        | 0.2115 | 1.71E-199 |
| ESO26      | 0.2114 | 2.39E-199 |
| PCI-38     | 0.2113 | 3.19E-199 |
| SK-PN-DW   | 0.2106 | 6.40E-198 |
| PSN1       | 0.2105 | 1.27E-197 |
| GI-1       | 0.2102 | 5.13E-197 |
| A375       | 0.2099 | 1.34E-196 |
| KP-N-YS    | 0.2092 | 3.20E-195 |
| A172       | 0.209  | 8.40E-195 |
| HuP-T4     | 0.2088 | 1.92E-194 |
| KP-3       | 0.2085 | 6.81E-194 |

|            |        |           |
|------------|--------|-----------|
| MKN28      | 0.2084 | 1.28E-193 |
| HCC2998    | 0.2084 | 1.27E-193 |
| EW-7       | 0.2081 | 3.94E-193 |
| TOV-112D   | 0.208  | 7.11E-193 |
| BB30-HNC   | 0.2078 | 1.48E-192 |
| HCC1143    | 0.2077 | 2.68E-192 |
| COR-L23    | 0.2075 | 4.84E-192 |
| MFE-296    | 0.2068 | 1.09E-190 |
| KYSE-220   | 0.2062 | 1.41E-189 |
| COLO-824   | 0.2058 | 8.79E-189 |
| AGS        | 0.2057 | 1.10E-188 |
| LC-1-sq    | 0.2056 | 2.29E-188 |
| NB7        | 0.2046 | 1.30E-186 |
| NCI-H358   | 0.2041 | 9.62E-186 |
| KP-1N      | 0.2036 | 1.02E-184 |
| NCI-H2170  | 0.2036 | 9.30E-185 |
| LS-123     | 0.2029 | 2.05E-183 |
| PCI-4B     | 0.2029 | 1.62E-183 |
| ES4        | 0.2024 | 1.24E-182 |
| A253       | 0.2023 | 1.76E-182 |
| HCC70      | 0.2017 | 2.44E-181 |
| SW1573     | 0.2014 | 9.13E-181 |
| ESS-1      | 0.2014 | 8.94E-181 |
| A2780cis   | 0.2014 | 8.89E-181 |
| HOP-62     | 0.2013 | 1.53E-180 |
| KYSE-150   | 0.201  | 5.65E-180 |
| D-502MG    | 0.2009 | 7.68E-180 |
| BxPC-3     | 0.2007 | 1.76E-179 |
| BE2-M17    | 0.2005 | 3.88E-179 |
| NCI-H23    | 0.1999 | 4.09E-178 |
| LS-180     | 0.1992 | 7.00E-177 |
| CAL-51     | 0.1987 | 7.43E-176 |
| LP-1       | 0.1983 | 3.98E-175 |
| TOV-21G    | 0.1982 | 4.68E-175 |
| TC-71      | 0.1981 | 7.97E-175 |
| GB-1       | 0.1949 | 3.50E-169 |
| KYSE-410   | 0.1948 | 4.81E-169 |
| LB771-HNC  | 0.1946 | 1.29E-168 |
| LS-411N    | 0.1945 | 1.44E-168 |
| SU-DHL-8   | 0.1944 | 3.02E-168 |
| PA-TU-8902 | 0.1942 | 6.27E-168 |
| SW837      | 0.1933 | 2.32E-166 |
| HT55       | 0.1932 | 2.57E-166 |

|              |        |           |
|--------------|--------|-----------|
| NB10         | 0.1929 | 1.12E-165 |
| OCI-AML2     | 0.1929 | 8.72E-166 |
| TE-9         | 0.1915 | 3.19E-163 |
| T84          | 0.1906 | 9.07E-162 |
| RPMI-8226    | 0.1906 | 8.39E-162 |
| U251         | 0.19   | 8.80E-161 |
| KYSE-140     | 0.1897 | 3.08E-160 |
| HCT-116      | 0.1896 | 4.99E-160 |
| Detroit562   | 0.1881 | 1.79E-157 |
| COLO-320-HSR | 0.1877 | 8.72E-157 |
| SNU-61       | 0.1877 | 7.25E-157 |
| DBTRG-05MG   | 0.1875 | 1.65E-156 |
| HuP-T3       | 0.1864 | 1.26E-154 |
| GAMG         | 0.186  | 5.31E-154 |
| GP5d         | 0.1855 | 3.11E-153 |
| PCI-6A       | 0.1851 | 1.68E-152 |
| MDA-MB-231   | 0.1846 | 1.05E-151 |
| A2780        | 0.1842 | 5.87E-151 |
| HGC-27       | 0.1837 | 3.33E-150 |
| OCUB-M       | 0.1836 | 4.35E-150 |
| SNU-C5       | 0.1829 | 6.57E-149 |
| MCAS         | 0.1826 | 1.75E-148 |
| no-10        | 0.1813 | 2.59E-146 |
| HCE-4        | 0.1805 | 5.87E-145 |
| HCC1395      | 0.1805 | 5.54E-145 |
| KYSE-520     | 0.1804 | 8.04E-145 |
| SW48         | 0.1795 | 2.27E-143 |
| HO-1-u-1     | 0.1791 | 9.88E-143 |
| HSC-3        | 0.1768 | 3.95E-139 |
| KYSE-510     | 0.1767 | 5.76E-139 |
| CL-11        | 0.1761 | 5.42E-138 |
| OVCAR-8      | 0.1752 | 1.54E-136 |
| PA-TU-8988T  | 0.1737 | 2.94E-134 |
| LU-65        | 0.1736 | 4.19E-134 |
| NB5          | 0.1728 | 6.86E-133 |
| NCI-H2023    | 0.1719 | 1.57E-131 |
| SNG-M        | 0.1711 | 2.88E-130 |
| 42-MG-BA     | 0.1708 | 8.80E-130 |
| MDA-MB-468   | 0.1698 | 3.02E-128 |
| RKO          | 0.1694 | 1.05E-127 |
| Ca9-22       | 0.1687 | 1.08E-126 |
| SCC-9        | 0.1676 | 4.43E-125 |
| KYSE-450     | 0.1654 | 9.22E-122 |

|            |        |           |
|------------|--------|-----------|
| MDA-MB-436 | 0.1653 | 1.44E-121 |
| SU-DHL-5   | 0.1646 | 1.45E-120 |
| BHY        | 0.1633 | 1.24E-118 |
| SK-MG-1    | 0.1632 | 1.57E-118 |
| HCC-15     | 0.1627 | 8.80E-118 |
| OVK-18     | 0.1622 | 4.51E-117 |
| SW620      | 0.1621 | 5.62E-117 |
| KYSE-270   | 0.1617 | 2.15E-116 |
| YH-13      | 0.1616 | 3.38E-116 |
| SNU-81     | 0.1615 | 3.71E-116 |
| JMT-1      | 0.161  | 2.29E-115 |
| EW-1       | 0.1603 | 1.95E-114 |
| KP-4       | 0.1552 | 3.23E-107 |
| RL95-2     | 0.1547 | 1.27E-106 |
| LoVo       | 0.1546 | 2.07E-106 |
| DAN-G      | 0.1514 | 4.43E-102 |
| HCC-78     | 0.1509 | 1.81E-101 |
| MFM-223    | 0.1482 | 6.21E-98  |
| NUGC-3     | 0.1439 | 2.32E-92  |
| JHU-029    | 0.1415 | 2.41E-89  |
| ES8        | 0.1371 | 6.77E-84  |
| PCI-30     | 0.1353 | 1.08E-81  |
| CAL-33     | 0.1338 | 5.04E-80  |
| FADU       | 0.1331 | 3.77E-79  |
| NCI-H1944  | 0.1327 | 1.23E-78  |
| MDST8      | 0.1258 | 8.70E-71  |
| HCT-15     | 0.1243 | 3.59E-69  |
| BICR78     | 0.1227 | 2.05E-67  |
| MDA-MB-453 | 0.1175 | 5.61E-62  |
| TE-10      | 0.1149 | 2.35E-59  |
| PC-14      | 0.1147 | 4.36E-59  |
| LCLC-97TM1 | 0.1146 | 5.23E-59  |

**Supplementary Table 8. 10 potential prognosis biomarkers for breast cancer screened by PES-BRCA.** P-value was calculated by the log-rank test (two-sided, n=1115).

| Protein | PES-BRCA | P-value | PMID                  |
|---------|----------|---------|-----------------------|
| RPF2    | 0.9814   | 0.0049  | NA                    |
| RPS24   | 0.9724   | 0.006   | NA                    |
| SDAD1   | 0.9639   | 0.0025  | NA                    |
| TCP1    | 0.9636   | 0.0049  | 37539774 <sup>4</sup> |
| MRPL13  | 0.9529   | 0.0021  | 33658859 <sup>5</sup> |
| GARS1   | 0.9480   | 0.00072 | NA                    |
| PSMD7   | 0.9454   | 0.0053  | 33042429 <sup>6</sup> |
| CCT6A   | 0.9427   | 0.0027  | 37190091 <sup>7</sup> |
| TARS1   | 0.9355   | 0.00027 | 37637061 <sup>8</sup> |
| RPS29   | 0.9265   | 0.00281 | 37198999 <sup>9</sup> |

**Supplementary Table 9. Top 10 microproteins with the highest PES predicted by PIC models**

| Microprotein | Length | hPES   | mPES   | cPES   | PES    |
|--------------|--------|--------|--------|--------|--------|
| smORF594570  | 77     | 0.9925 | 0.9477 | 0.9836 | 0.9746 |
| smORF594569  | 59     | 0.9868 | 0.9243 | 0.9608 | 0.9573 |
| smORF597959  | 100    | 0.9870 | 0.9133 | 0.9234 | 0.9413 |
| smORF597276  | 99     | 0.9137 | 0.9320 | 0.9666 | 0.9374 |
| smORF614125  | 94     | 0.8883 | 0.9295 | 0.9791 | 0.9323 |
| smORF614698  | 97     | 0.8920 | 0.9253 | 0.9545 | 0.9239 |
| smORF605028  | 93     | 0.9245 | 0.8776 | 0.9566 | 0.9196 |
| smORF598483  | 79     | 0.9846 | 0.8173 | 0.9458 | 0.9159 |
| smORF590947  | 52     | 0.9467 | 0.8483 | 0.9437 | 0.9129 |
| smORF587640  | 81     | 0.9788 | 0.7761 | 0.9535 | 0.9028 |

**Supplementary Table 10. AUROC of models trained with different sequence representation methods.** The best performance values were highlighted in bold. The AUROC of cell-level models was calculated on 323 cell line models. **One-hot\***: One-hot is a representation method that converts categorical data, such as amino acids or nucleotides in a sequence, into binary vectors where each position corresponds to a specific category, and only one position is set to 1 while all others are set to 0. **K-mer\***: K-mer is a sequence representation method that breaks down a biological sequence into subsequences of length K, capturing local sequence patterns and features. **PSSM\***: Position-Specific Scoring Matrix (PSSM) is a representation method that captures the conservation of amino acids at each position in a multiple sequence alignment by scoring the likelihood of each amino acid occurring at each position relative to a background distribution. #: Other pre-trained protein language models.

| Methods                 | Human-level   | Cell-level<br>( $\bar{x} \pm \text{std}$ ) | Mouse-level   |
|-------------------------|---------------|--------------------------------------------|---------------|
| <b>ESM-2</b>            | <b>0.9411</b> | <b>0.8712<math>\pm</math>0.024</b>         | <b>0.9027</b> |
| ProtT5 <sup>#</sup>     | 0.9139        | 0.8466 $\pm$ 0.021                         | 0.8314        |
| ProtBert <sup>#</sup>   | 0.8878        | 0.8373 $\pm$ 0.019                         | 0.8337        |
| ProtAlbert <sup>#</sup> | 0.9180        | 0.7872 $\pm$ 0.027                         | 0.7562        |
| ProtXLNet <sup>#</sup>  | 0.8086        | 0.7401 $\pm$ 0.018                         | 0.6730        |
| One-hot*                | 0.8194        | 0.7335 $\pm$ 0.018                         | 0.6576        |
| K-mer*                  | 0.9070        | 0.7223 $\pm$ 0.024                         | 0.6771        |
| PSSM*                   | 0.8433        | 0.6419 $\pm$ 0.024                         | 0.5171        |

**Supplementary Table 11. AUROC and runtime required for feature extraction time of models with different parameter sizes.** The best performance values were highlighted in bold.

The AUROC of cell-level models was calculated on 323 cell line models.

| Parameter      | Human-level |               | Cell-level |                                       | Mouse-level |               |
|----------------|-------------|---------------|------------|---------------------------------------|-------------|---------------|
|                | Time(h)     | AUROC         | Time(h)    | AUROC<br>( $\bar{x} \pm \text{std}$ ) | Time(h)     | AUROC         |
| t48_15B_UR50D  | 13.7        | <b>0.9411</b> | 4.53       | 0.8514 $\pm$ 0.02                     | 2.05        | 0.8050        |
| t36_3B_UR50D   | 3.63        | 0.9369        | 1.55       | 0.8475 $\pm$ 0.03                     | 0.48        | 0.8294        |
| t33_650M_UR50D | 0.92        | 0.9070        | 0.24       | <b>0.8712<math>\pm</math>0.02</b>     | 0.13        | <b>0.9027</b> |
| t30_150M_UR50D | 0.27        | 0.8743        | 0.08       | 0.8487 $\pm$ 0.02                     | 0.06        | 0.8566        |
| t12_35M_UR50D  | 0.13        | 0.8680        | 0.04       | 0.8313 $\pm$ 0.02                     | 0.02        | 0.7752        |
| t6_8M_UR50D    | 0.08        | 0.8499        | 0.03       | 0.8142 $\pm$ 0.02                     | 0.01        | 0.7608        |

**Supplementary Table 12. AUROC of models trained with different pooling strategy.** The best performance values were highlighted in bold. The AUROC of cell-level models was calculated on 323 cell line models. **CLS\***: CLS pooling is a strategy that uses the special [CLS] token's embedding from the output of a transformer model to represent the entire sequence. **Average\***: Average pooling is a strategy that computes the average of all token embeddings in a sequence to represent the entire sequence. **Max\***: Max pooling is a strategy that selects the maximum value from each dimension of the token embeddings in a sequence to represent the entire sequence.

| ID | CLS* | Mean* | Max* | Human-level   | Cell-level<br>( $\bar{x} \pm \text{std}$ ) | Mouse-level   |
|----|------|-------|------|---------------|--------------------------------------------|---------------|
| 1  | √    |       |      | 0.8631        | 0.8557 $\pm$ 0.019                         | 0.8585        |
| 2  |      | √     |      | <b>0.9036</b> | <b>0.8664<math>\pm</math>0.022</b>         | 0.8659        |
| 3  |      |       | √    | 0.8615        | 0.8491 $\pm$ 0.018                         | 0.8606        |
| 4  | √    | √     |      | 0.8904        | 0.8613 $\pm$ 0.021                         | 0.8624        |
| 5  | √    |       | √    | 0.8674        | 0.8559 $\pm$ 0.017                         | <b>0.8701</b> |
| 6  |      | √     | √    | 0.8648        | 0.8620 $\pm$ 0.019                         | 0.8533        |
| 7  | √    | √     | √    | 0.8710        | 0.8612 $\pm$ 0.021                         | 0.8239        |

**Supplementary Table 13. Hyperparameter combinations used in the grid search.**

| Hyperparameters | Search space          |
|-----------------|-----------------------|
| Epoch           | [5,10,15,20]          |
| Hidden size     | [160,320,640,1280]    |
| Learning rate   | [1e-3,1e-4,1e-5,1e-6] |
| Batch size      | [32,64,128,256]       |

**Supplementary Table 14. Optimal combination of hyperparameters for the best model performance.** The AUROC of cell-level models was calculated on 323 cell line models.

| Level                                | AUROC        | Epoch | Hidden size | Learning rate | Batch size |
|--------------------------------------|--------------|-------|-------------|---------------|------------|
| Human                                | 0.9366       | 20    | 1280        | 1e-4          | 256        |
| Cell<br>( $\bar{x} \pm \text{std}$ ) | 0.8676±0.015 | 20    | 640         | 1e-5          | 128        |
| Mouse                                | 0.9127       | 20    | 320         | 1e-5          | 64         |

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